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The Immunobiology of Transplantation

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CHAPTER OUTLINE

KEY POINTS

- Allograft rejection occurs because the recipient's immune system recognizes the donor's human leukocyte antigens (HLA) as non-self through three pathways. In the direct pathway, recipient CD8+ T cells recognize intact donor HLA class I molecules on donor antigen presenting cells (APC) that migrate from the allograft to the recipient's lymph nodes. In the indirect pathway, donor peptides are processed by the recipient's APC and expressed on HLA class II molecules, where they are recognized by recipient CD4+ T lymphocytes. In the semidirect pathway, donor HLA molecules are transferred on recipient APC that can then interact with recipient CD8+ T lymphocytes.
- After alloantigen recognition by the T-cell receptor, cosignals must take place for T-cell activation to occur. Although many other cosignals exist, the interaction between B7 at the surface of APC and CD28 on the T-cell surface is one of the important stimulatory cosignals, while the interaction between B7 and CTLA4 is an important coinhibitory pathway.
- Following their activation and depending on signals provided by the inflammatory milieu, CD4 T cells can differentiate into at least five major subsets: Th1, Th2, Th17, follicular helper T (Tfh) cells, and regulatory T (Treg) cells. Three subtypes are involved in immunogenic responses and rejection: Th1, Th17, and Tfh. Two subtypes display tolerogenic roles: Th2 and Treg.
- B lymphocytes become activated when their B-cell receptor engages an alloantigen (soluble or presented by an APC) that they internalize, process, and re-express on HLA class II molecules. Through interactions with cognate CD4+ follicular helper T cells, B lymphocytes differentiate into antibody-producing plasmocytes or memory B cells.
- Important effector mechanisms of allograft rejection include T-lymphocyte (CD8+) cytotoxicity through the perforin-granzyme and the Fas/Fas-ligand pathways and donor-specific, anti-HLA antibody-mediated damage, which is secondary to complement activation through the classical pathway or antibody-dependent cytotoxicity.

Allograft rejection occurs because antigenic determinants present within the allograft are recognized by the recipient's immune system as non-self, dangerous, or both. In this chapter, we will first review the mechanisms involved in allorecognition. Then, we will discuss transplant-relevant antigens and antibodies. Finally, we will provide a mechanistic overview of acute kidney graft rejection and immunosuppressive drugs. The clinical aspects of acute kidney graft rejection are covered in greater depth in Chapter 70.

MECHANISMS INVOLVED IN ALLORECOGNITION

T cells recognize foreign human leukocyte antigens (HLAs) through their T-cell receptors (TCR) after these antigens are presented by APCs. The TCR is expressed at the cell surface in conjunction with CD3, which is present on all mature T cells. Both cytotoxic (CD8+) and helper (CD4+) T cells are activated by the CD3-TCR complex. Different cells within the immune system, such as dendritic cells (DCs), monocytes, macrophages, and B cells, play important roles in presenting antigens to T cells. In 95% of cases, the TCR is composed of α and β chains while in 5% of cases, it is made up of γ and δ chains. The TCR is linked to the cell membrane CD3 complex, which comprises CD3ε/γ and CD3ε,/δ with a homodimer of a ζ-chain (Fig. 69.1). Upon TCR ligation, phosphorylation of the CD3 complex favors calcium release from the endoplasmic reticulum and calcium binding to calmodulin, leading to activation of the phosphatase activity of calcineurin. Calcineurin dephosphorylates nuclear factor of activated T cells (NFAT), allowing for its nuclear translocation and transcription of a series of effector genes, including interleukin-2 (IL-2) and the IL-2 receptor (IL-2R). Two major T cell subsets, CD4 and CD8 T cells, are central players in allograft rejection.

Antigen presentation is an integral component of the normal immune response that helps protect against infectious agents and maladapted cells, such as cancer cells. However, in transplantation, the immune system is activated by alloimmune triggers, that is, antigens that are genetically encoded to differ between two individuals of the same species, such as the HLA molecules, which are encoded by the major histocompatibility complex (MHC). Antigen presentation may follow one of two major pathways: direct antigens presentation by MHC class I (MHC I) molecules and indirect antigens presentation by MHC II molecules.

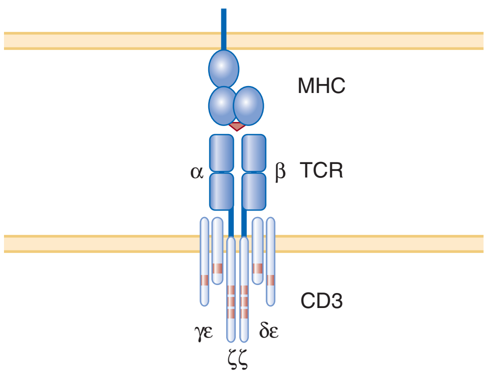

Fig. 69.1 The T-cell receptor. T-cell receptors (TCR) alpha/beta heterodimers associate with the chains of the CD3 complex. This association is essential for expression of the TCR on the cell surface. The CD3 chains are also essential for signal transduction. The red dots represent immunoreceptor tyrosine-based activation motifs that can be phosphorylated when the TCR recognizes a peptide presented by a major histocompatibility complex molecule. (Permission needed from: Stauss H, et al. Monoclonal T cell receptors: New reagents for cancer therapy. *Molecular Therapy*. 2007;15(10):1744–1750.)

ANTIGEN PRESENTATION PATHWAYS

DIRECT PATHWAY

In the transplantation setting, intact MHC I molecules expressed on donor cells can be recognized by alloreactive recipient T cells. This mode of recognition is referred to as direct allorecognition. Direct allorecognition[1,2] is thought to depend on the presence of donor DCs that traffic from the allograft to secondary lymph nodes, where they interact with recipient CD8 cytotoxic T cells. Large numbers of alloreactive T cells are present in all individuals, including transplant recipients, since thymic maturation of T cells deletes only autoreactive T cells without affecting alloreactive T cells.[3] The direct response is thought to be rapid and relatively short-lived owing to eventual destruction of donor DCs.[4]

MHC I molecules are expressed on all nucleated cells. In the normal nontransplanted host, MHC I molecules at the surface of APCs also interact with CD8 T cells. However, in the normal host, peptide antigens bound to MHC I molecules are being recognized rather than the MHC molecules themselves. Antigenic peptides presented by MHC I (HLA-A, B, C) usually originate from endogenous proteins that have been degraded to 8 or 9 amino acid peptides by the proteasome complex.[5] Proteasomes are composed of a barrel-shaped catalytic 20S complex capped at both ends by 19s regulatory complexes that control access to the catalytic core. Proteins that are damaged or dysfunctional are targeted for degradation by the addition of ubiquitin tags in a process called ubiquitination. Polyubiquitinated proteins are recognized by the 19S complex, allowing them to move toward the proteolytic core for degradation. This process is responsible for degradation of 75% to 95% of intracellular proteins.[6] Peptides are degraded down to 8 or 9 residues and loaded onto MHC I molecules in the endoplasmic reticulum by the transporter associated with antigen processing (TAP).[7] In the nontransplant setting, presentation of antigenic peptides by MHC I molecules plays an important role in the immunogenicity of mutant sequences with regard to cancer cells or proteins that stem from a viral infection. In the setting of transplantation, however, donor peptides presented within the MHC I groove of donor APCs can impact, either positively or negatively, the capacity to present allogeneic determinants to the recipient's immune system. They can alter the stability of the immunologic synapse (the site of physical interaction between APC and T cell), and therefore modulate the robustness of the response.

INDIRECT PATHWAY

The indirect presentation pathway proceeds when allopeptides (fragments of donor HLA molecules) are loaded onto the

MHC II molecules expressed on the recipient's APCs. Non-MHC peptides can also be presented in this manner since, in the nontransplanted host, this pathway deals with the presentation of endocytosed antigens digested by the lysosomal pathway, which are re-expressed on the cell surface as peptides (≥15 amino acids) bound to MHC II molecules. MHC II-bound peptides are recognized by the TCR of CD4 T cells, leading to their activation when proper costimulatory signals are also provided. MHC II molecules are expressed on a restricted number of immune cells, such as DCs, monocytes, macrophages, and B cells. However, interferon-γ can induce the expression of MHC II on several cell types, including epithelial and endothelial cells, which can then acquire non-professional antigen-presenting capacity.

MHC II molecules are formed in the endoplasmic reticulum, where they associate with the invariant chain ''li'' that fills the peptide-binding groove.[8] The invariant chain targets MHC II for the endosomal pathway, where various degradation steps allow for the loading of MHC II with antigenic peptides. Once loaded and stabilized, the MHC II molecule moves to the plasma membrane. In the transplant setting, peptides derived from polymorphic regions of the donor MHC molecules can be taken up by recipient APCs, degraded within the endocytic compartment, and loaded onto recipient MHC II molecules for presentation to alloreactive recipient CD4 T cells. MHC II molecules expressed by DCs play an important role in initiating the indirect antigen presentation pathway through interactions with CD4 T cells. The indirect pathway is thought to be activated in the long term after transplantation and may play an important role in promoting alloantibody production and chronic antibody-mediated allograft rejection.

SEMIDIRECT PATHWAY AND CROSS-DRESSING

The semidirect pathway was described more recently, and refers to the transfer of intact donor MHC molecules to the recipient's APCs.[9–12] The semidirect pathway is posited to play an important role in T-cell allosensitization after donor APCs have been eliminated by the host response, or when these cells are unable to migrate to secondary lymph nodes because of severed lymphatic vessels at the time of transplantation. In the semidirect pathway, intact donor HLA molecules can be picked up by recipient APCs and re-expressed on their surface, a mechanism termed *cross-dressing*. Directly alloreactive recipient T cells can then interact with cross-dressed APCs, leading to their activation.

The semidirect pathway helps explain the activation of naïve T cells in situations where donor DCs are not available for alloantigen presentation. In several transplantation models, donor DCs have been detected only at very low numbers in draining lymphoid organs, and they have been shown to be rapidly eliminated by natural killer (NK) cells.[13–15] Recipient DCs rapidly invade and replace donor DCs within a few days after transplantation. It is thought that the semidirect pathway is used by recipient DCs, in conjunction with the indirect pathway, to present alloantigens to naïve recipient T cells, explaining the long-term capacity to mount a rejection response even after donor DCs have been destroyed. Although different types of DCs have been implicated in alloantigen presentation, monocyte-derived DCs are thought to play a central role in the sensing of self and non-self antigens, and in the presentation of donor alloantigens through the semidirect pathway.[4,16] The mechanisms responsible for cross-dressing of donor MHC molecules on DCs are the subject of intense investigation. It was previously thought that MHC molecules shed from the allograft could be taken up by DCs and re-expressed on the cell surface. More recently, a potential role for extracellular vesicles in the transfer of intact donor MHC molecules was suggested. All cell types can release a wide array of extracellular vesicles that can serve as intercellular cargo for protein, mRNA, and microRNA. Exosomes—extracellular vesicles of endocytic origin—can carry MHC molecules and are thought to initiate cross-dressing of MHC molecules when engulfed by DCs.[17,18]

The antigen presentation capacity of DCs (whether allogenic or not) varies with their state of maturation and activation. Release of danger-associated molecular patterns (DAMPs) by the graft, in association with ischemia-reperfusion at the time of transplantation, favors maturation of DCs.[4] Exosomes released by the allograft can also promote inflammation and DC maturation.[19,20] Regardless of donor or recipient origin, DCs migrate to secondary lymphoid tissue to interact with naïve T cells. However, under certain inflammatory conditions, such as chronically rejecting allografts, structures reminiscent of lymphoid organs can form within allografts. These structures, called tertiary lymphoid organs, allow antigen presentation within the allograft, thus favoring the production of allo- and autoantibodies[19,21] that are important in rejection.

T-CELL COSIGNALING PATHWAYS

The TCR is an octomeric complex of variable TCR receptor α and β chains with three dimeric signaling modules^{CD3}δ/ε, CD3 γ/ε and^{CD247}ζ/ζ or ζ/η (Fig. 69.1). TCR ligation activates Signal 1. However, a second signaling pathway, Signal 2, needs to be triggered for T-cell activation and acquisition of effector function. Pathways that activate or inhibit Signal 2 are known as cosignaling pathways (Fig. 69.2).[22] These pathways have profound implications on T-cell responses to antigenic determinants, controlling activation, proliferation, differentiation, cytokine production, and effector functions.

The first and most well-characterized costimulatory pathway depends on interactions between the CD28 receptor, expressed on naïve T cells, and B7 ligands, expressed on APCs.[23–28] CD28 is constitutively expressed by naïve CD4 and CD8 T cells, and, upon ligation with B7.1 (CD80) or B7.2 (CD86), it activates cosignaling pathways that drive activation and survival of T cells. The expression of B7.1 and B7.2 by APCs is influenced by their activation state, which can be triggered by exposure to proinflammatory cytokines, such as interleukin-1β (IL-1β), interleukin-6 (IL-6), and tumor necrosis factor-alpha (TNF-α). Cytotoxic T-lymphocyte antigen-4 (CTLA4) was the first coinhibitory pathway to be described.[29–32] CTLA4 overexpression and concomitant downregulation of CD28 by endocytosis help prevent unregulated T-cell activation. Since the initial description of co-signaling pathways in the late 1980s, more than 25 different types of cosignaling pathways have been characterized. Most belong to the immunoglobulin superfamily (IgSF) and the tumor necrosis factor receptor superfamily (TNFRSF). Other costimulatory pathways that have emerged as important regulators of T-cell activation in transplantation include the CD40/CD154 (CD40 ligand) and CD278/CD275 pathways. CD40, a member of the TNFRSF expressed on most APCs, interacts with the CD40 ligand expressed on T cells, favoring their activation. This pathway impacts both T-cell activation and T- and B-cell interactions. Inducible T-cell costimulator (ICOS or CD278), a member of the IgSF, is expressed on activated T cells, and favors further proliferation, cytokine production, and effector functions upon interaction with CD275.[33]

T-CELL PHENOTYPES

Following their activation and depending on signals provided by the inflammatory milieu, CD4 T cells can differentiate into at least five major subsets: Th1, Th2, Th17, follicular helper T (Tfh) cells, and regulatory T (Treg) cells. Three subtypes are involved in immunogenic responses and rejection: Th1, Th17, and Tfh. Two subtypes display tolerogenic roles: Th2 and Treg.

Th1 cells have long been considered the cornerstone of allograft rejection. Activated Th1 CD4 T cells produce IL-2 and express IL-2R, with their interaction prompting proliferation. The transcription factor T-bet, also present in Th1 cells, favors the expression of interferon-γ and TNF-α. Cytokines released by Th1 CD4 T cells promote the activation of CD8 cytotoxic T cells, which in turn recognize allograft cells through the direct pathway and use the perforin-granzyme cytolytic pathway to destroy them. Interferon-γ produced by Th1 cells also induces the production of complement-activating antibodies by B cells, therefore recruiting an additional mechanism of allograft rejection.

Th17 cells are induced under the combined actions of transforming growth factor-β (TGF-β), IL-6, and IL-1β. They produce interleukin-17 (Il-17), interleukin-21 (IL-21), and interleukin-22 (IL-22) following the activation of the transcription factors STAT3 and RORγt. Th17 cells have been implicated in many autoimmune diseases in addition to transplant rejection.[34]

Tfh cells secrete interleukin-4 (IL-4) and IL-21, and express CD40 ligand. The interaction between CD40 ligand and CD40, expressed on B cells, promotes the maturation of naïve B cells into memory B cells and antibody-secreting plasmocytes.

Th2 cells are considered anti-inflammatory and are characterized by the production of IL-4, interleukin-5 (IL-5), and interleukin-13 (IL-13). IL-4 produced by Th2 cells induces the differentiation of CD8 T cells into a noncytotoxic phenotype, and the production of non-complement fixing antibodies by B cells. Th2 cytokines can also prompt the emergence of Treg cells.

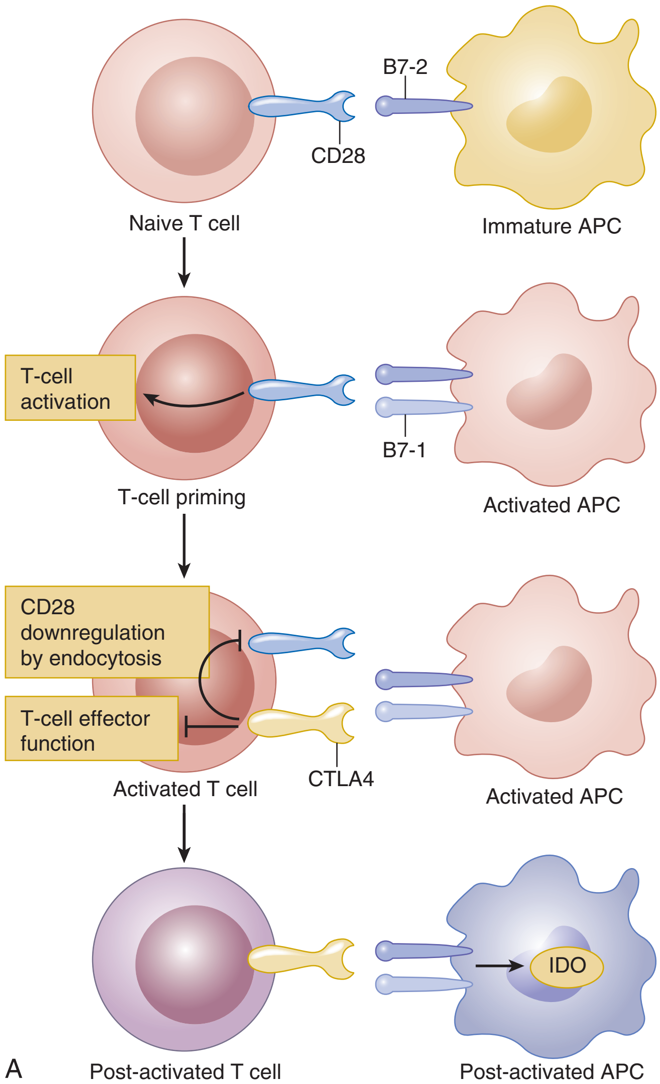

Fig. 69.2 Cosignaling interactions in T cells. A, CD28 is constitutively expressed on the cell surface of naive CD4+ and CD8+ T cells and provides an essential costimulatory signal for T-cell growth and survival upon ligation by B7-1 and B7-2 on antigen-presenting cells (APCs). Cytotoxic T lymphocyte antigen 4 (CTLA4) is induced, following T-cell activation, and it suppresses T-cell responses. When CTLA4 is upregulated, CD28 expression is subsequently downregulated by endocytosis. Expression of B7-1 and B7-2 is modulated by the activation state of the APC. B7-2 is constitutively expressed on APCs at low levels, and infection, stress, and cellular damage recognition by innate receptors activate APCs and induce transcription, translation, and transportation of both B7-1 and B7-2 to the cell surface. *Continued*

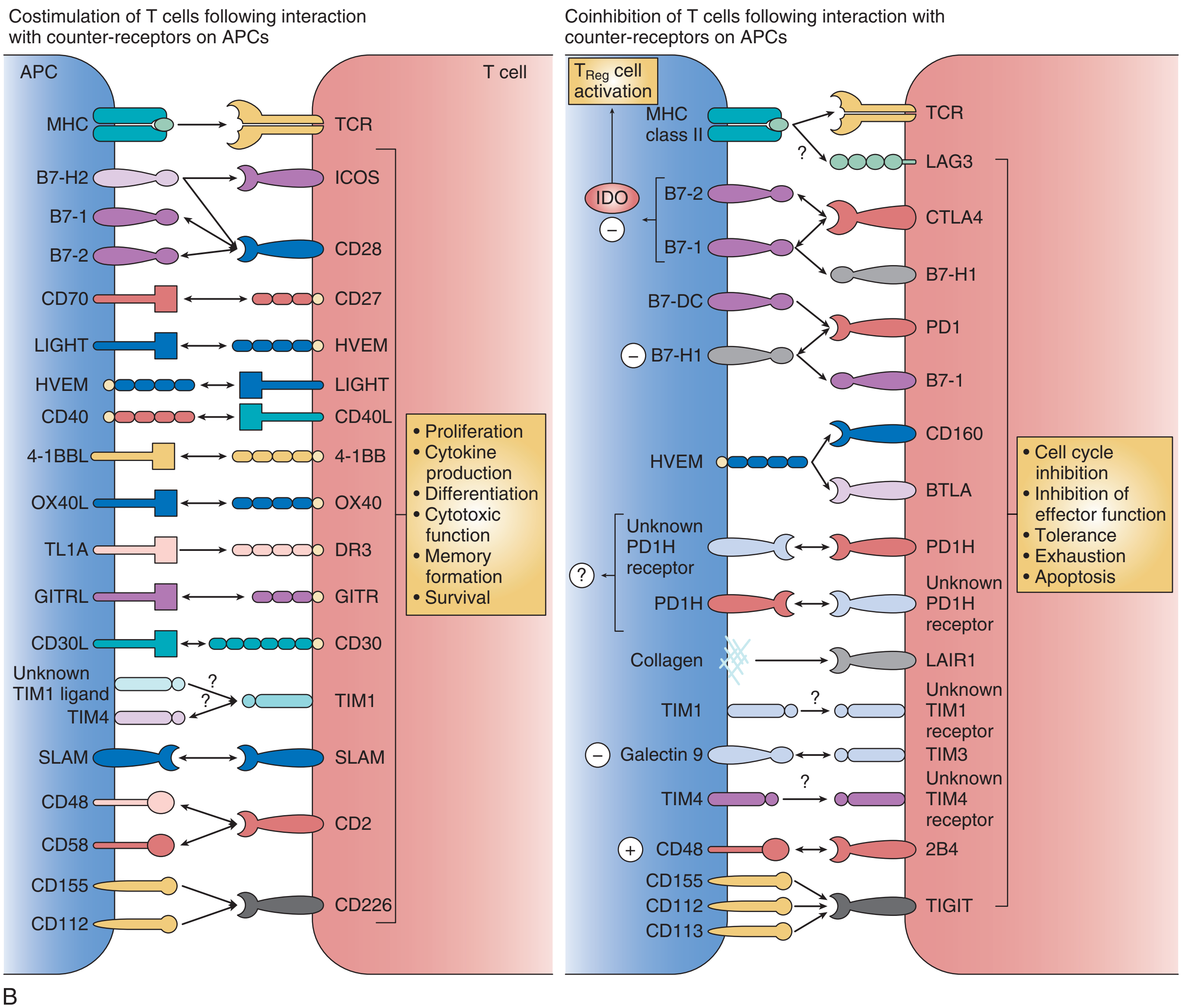

Fig. 69.2, cont'd B, *Left panel*, Costimulatory molecules deliver positive signals to T cells following their engagement by ligands and counter-receptors on APCs. Several costimulatory molecule interactions are bidirectional. *Right panel*, Coinhibitory molecules deliver negative signals into T cells. CTLA4 is involved in bidirectional interactions: It inhibits T-cell function after binding B7-1 and B7-2, and CTLA4-bound B7-1 and B7-2 may induce the expression of indoleamine 2,3-dioxygenase (IDO), which acts in *trans* to suppress activation of conventional T cells and promote the function of regulatory T cells. (From Chen L, Flies DB. Molecular mechanisms of T cell co-stimulation and co-inhibition. *Nat Rev Immunol*. 2013;13(4):227–242. With permission.)

Treg cells express CD4, CD25, and the transcription factor forkhead box P3 (FOXP3). They are considered the primary mediators of peripheral tolerance. Like effector T cells, Treg cells express a TCR that activates calcineurin-dependent pathways upon ligation with antigenic peptides presented by MHC II molecules. Their activation is dependent on two signals (Fig. 69.3).[35] However, unlike effector T cells, the second signal does not depend on interactions between CD28 and B7.1 or B7.2, since CTLA4 expressed on Treg cells inhibits CD28 signaling. Rather, the second signal is initiated by IL-2/IL-2R interactions, and thus tends to occur in the presence of high levels of IL-2. Treg cells have been shown to suppress the function of CD4 T cells, CD8 T cells, B cells, and macrophages. Their suppressive activity can inhibit a number of immunogenic reactions, including allograft rejection. This immunosuppressive function depends on at least 4 mechanisms of action:

- Secretion of inhibitory cytokines such as interleukin-10 (IL-10), TGF-β, and interleukin-35 (IL-35)
- Suppression by metabolic disruption: Treg cells express the ectoenzymes CD39 and CD73, therefore favoring extracellular ATP and ADP metabolism to adenosine. The latter interacts with the A2A adenosine receptor on effector T cells, thereby launching signaling pathways that inhibit effector T-cell activation[36]

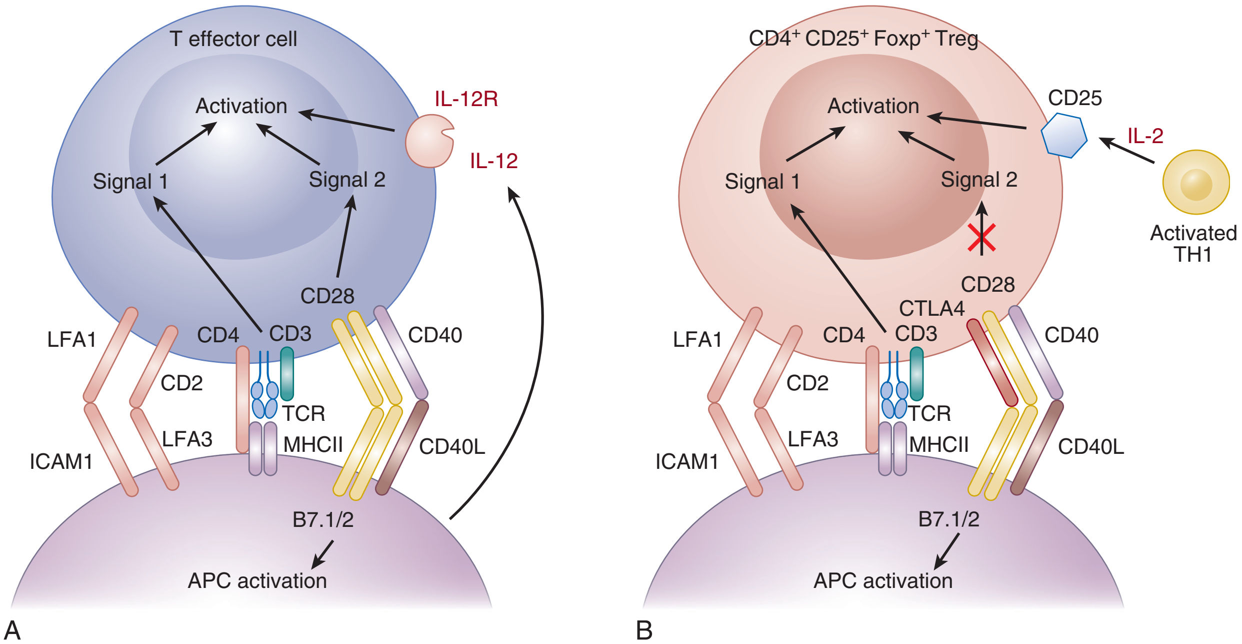

Fig. 69.3 Activation of effector and regulatory T cells by antigen-presenting cells. Key surface molecules in activation of A, T effector cells and B, T regulatory cells (Tregs). The key molecules required for both cells are similar. The T-cell receptor complex includes CD3, CD2, CD4 or CD8, LFA1, and CD45R, and activation of T-cell receptor (TCR) by antigen results in Signal 1 for T effector cells and Tregs. In effector T cell-lineage T cells, CD28 on the T cells is activated by B7.1 and B7.2 on antigen-presenting cells (APCs) and generates Signal 2, which, combined with Signal 1, initiates effector T-cell activation. The activation of effector T cells is augmented by CD40L binding to CD40 and cytokines, such as IL-2 and IL-12, for generation of Th1 cells. With Tregs, CTLA4 binds to B7.1 and B7.2 and limits activation through CD28. Thus the effector T cells Signal 2 pathway is not required for Treg activation. The second signal for Treg activation is generated by IL-2 binding to the IL-2 receptor, which includes CD25. (From Hall BM. T cells: soldiers and spies—the surveillance and control of effector T cells by regulatory T cells. *Clin J Am Soc Nephrol*. 2015;10(11):2050–2064. With permission.)

- Suppression by cytotoxicity: Treg cells employ the granzyme-perforin lytic system to kill activated NK cells and cytotoxic CD8 T cells
- Modulation of the activation and maturation of APCs, such as DCs

Following their activation, CD4 T cells initiate a proliferative response regulated by the mammalian target of rapamycin (mTOR) pathway, thus leading to clonal expansion. Depending on their phenotype, CD4 T cells can favor either activation or inhibition of other components of the adaptive immune response, including CD8 T cells and B cells. Activation of cytotoxic CD8 T cells can, in turn, directly damage and kill allograft cells through the release of perforin and granzyme B, whereas maturation of B cells into antibody-secreting plasmocytes can lead to the production of donor-specific antibodies and autoantibodies that are detrimental to the allograft.

B CELLS AND ANTIBODY PRODUCTION

The humoral response, mediated by antibody-producing B cells, and leading to antibody-dependent damage to the allograft, is the other major component of the adaptive immune system that is of pivotal importance in allograft rejection. Hyperacute antibody-mediated rejection was described in the 1960s as an extremely severe and rapid form of rejection dependent on the presence of preformed anti-HLA antibodies that target the allograft endothelium as soon as reperfusion is established. The use of crossmatching to screen for the presence of preformed anti-HLA antibodies has resulted in a sharp decrease in the occurrence of hyperacute antibody-mediated rejection. Hence the importance of antibodies and B cells as mediators of rejection was forgotten or dismissed until the turn of the 21st century, when standardization of C4d staining for kidney allograft biopsies—a readout of complement activation by immune complexes—reactivated the field of antibody-dependent rejection. HLA typing and anti-HLA antibodies are discussed in later in this chapter.

B cells exert a number of important immunologic functions; the most studied is the production of antibodies that can bind protein, peptide, and carbohydrate antigens (as opposed to T cells that react exclusively to antigenic peptides). B cells are good APCs and play an important role in the architecture of secondary lymphoid organs. They are characterized by the surface expression of a B-cell receptor (BCR), composed of a membrane-bound immunoglobulin (Ig) molecule associated with an Ig αβ heterodimer (CD79a and CD79b) that controls downstream signaling. B cells also express different surface markers depending on their state of maturation and differentiation. The B-cell lineage contains three major subsets: B1, B2, and regulatory B cells. (Fig. 69.4)[37] B1 cells are found in the peritoneal and pleural cavity. They express CD19 and CD5, produce low-affinity polyreactive natural antibodies, and do not require T-cell help. B2 cells are formed in the bone marrow and circulate to the spleen and lymph

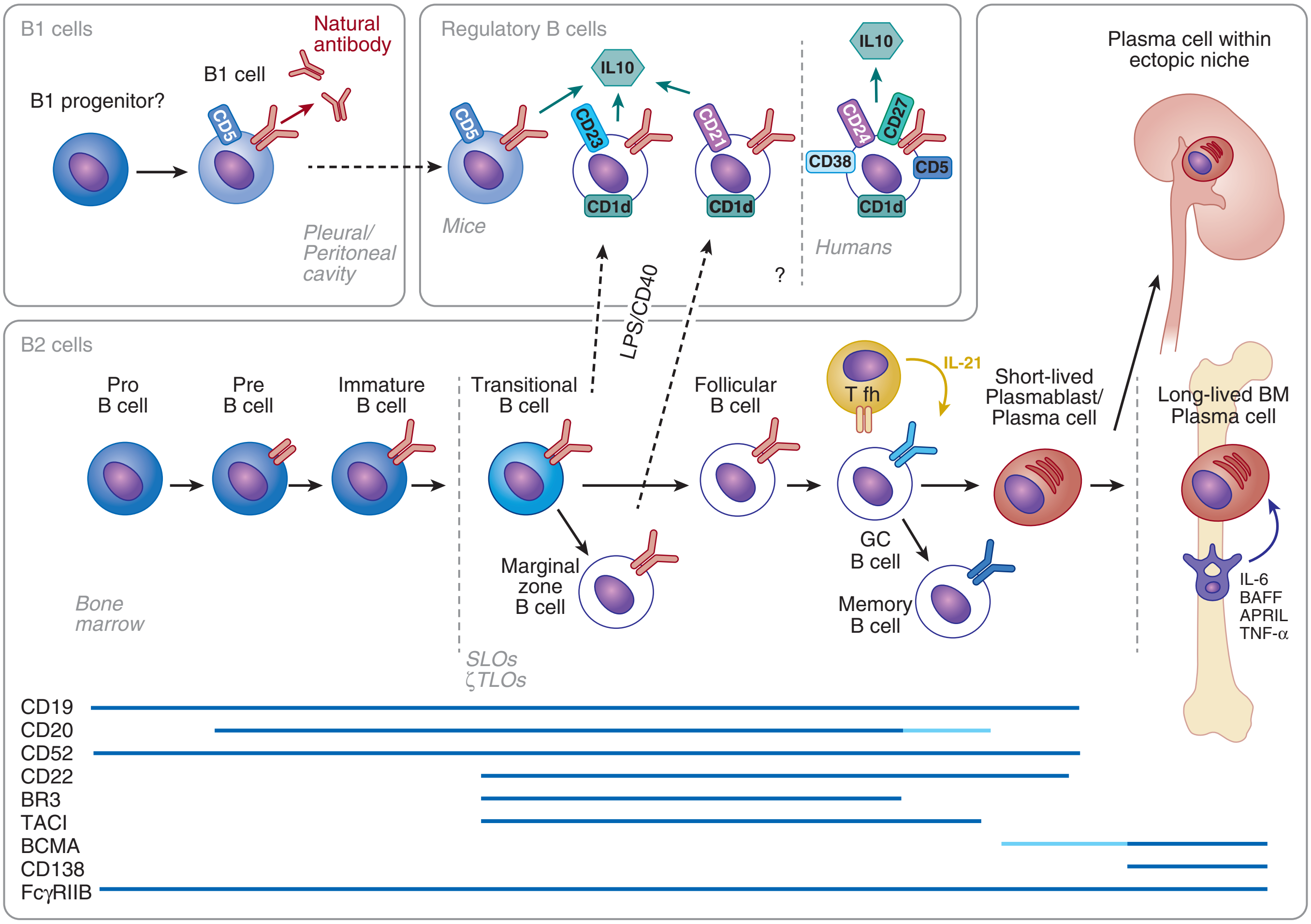

Fig. 69.4 B-cell ontogeny and differentiation. Most peripheral B cells are produced in the bone marrow and referred to as B2 cells. A minor B-cell population, known as B1 cells, are found not only in the pleural and peritoneal cavities but also in small numbers in the spleen. B1 cells express CD19, high levels of CD5, and produce low-affinity natural antibody (mainly IgM), without T-cell help. It is currently unclear whether B1 cells arise from a unique progenitor or from a progenitor common to both B1 and B2 cells. B2 cells are formed in the bone marrow and develop from pro-B cells, to pre-B cells, to immature B cells, which are released into the periphery. Following antigen encounter, B cells obtain T-cell help and enter the germinal center. Here they undergo class switch recombination and affinity maturation, a process involving iterative cycles of somatic hypermutation and proliferation. A specific subset of T cells, known as T follicular helper cells (Tfh) are critical for germinal center formation and are thought to provide both contact and cytokine (IL-21) signals for germinal center B cells. Short-lived plasmablasts and memory cells arise from GC B cells. Some plasmablasts circulate (depending on the expression of CXCR3 and CXCR4), and a small proportion find a suitable niche for long-term survival within bone marrow and inflamed tissue, for example, rejecting allografts. A variety of molecules are expressed by B2 cells during their maturation and activation, as shown; Fcγ RIIB is expressed throughout development, whereas most other markers are expressed either on B cells or on antibody-producing plasma cells. Dark-blue bars represent high expression, light-blue bars intermediate expression. Regulatory B cells are a recently described subset, which inhibits T-cell responses via the production of IL10. These cells are likely to arise from multiple B-cell subsets (B1, transitional and marginal zone). Regulatory B cells have recently been described in humans and are characterized by surface expression of CD5, CD1d, CD24, CD27, and CD38. There are also thought to be regulatory B cells, which act independently of IL-10 (not shown). (From Clatworthy MR. Targeting B cells and antibody in transplantation. *Am J Transplant*. 2011;11(7):1359–1367. With permission.)

nodes as immature B cells. They differentiate into either plasmablasts, memory B cells, or long-lived plasma cells following antigenic interaction and maturation within lymph nodes. Regulatory B cells with immune inhibitory functions are characterized by IL-10 secretion and the surface expression of CD24, CD38, CD5, and CD1d. Regulatory B cells represent approximately 5% of circulating B cells.

B cells become activated when their BCR engages an antigen, either soluble or presented by an APC, such as follicular DCs. This leads to phosphorylation of tyrosine residues within the cytoplasmic domains of CD79a and CD79b, and activation of downstream signaling pathways. Several receptors present on B cells, such as CD19 and toll-like receptors, or the presence of complement components, such as C3b and C3d, can lower the threshold of signaling initiation. The presence of cytokines, such as B-cell activating factor (BAFF), can also enhance B-cell activation and survival. Full B-cell activation and maturation occurs in the lymph nodes, which are composed of an outer portion or cortex of B-cell lymphoid follicles, and an inner portion rich in immature and mature T cells. Upon antigen recognition by the BCR, B cells internalize and process bound antigen, and re-express antigenic peptides on MHC II molecules. These B cells then move to the border between the follicle and the T-cell zone, where they present antigen to a cognate (i.e., matching) Tfh CD4 T cell. This first phase of activation can result in the differentiation of B cells into short-lived plasmablasts that produce low-affinity IgM antibodies, or the formation of a germinal center that will allow B-cell differentiation into either memory B cells or long-lived plasma cells that are responsible for the production of high-affinity antibodies. (Fig. 69.5)[38] Germinal centers are formed when B cells proliferate after BCR engagement, forcing naive B cells to leave the follicle. B cells then present MHC II-bound antigens, in association with CD40 and CD80 or CD86, to Tfh cells for a second time. This leads to interactions with the TCR, CD40 ligand, and CD28 on follicular T cells. In addition, follicular T cells produce IL-21, which favors B-cell clonal expansion and antibody class switch recombination.

Antibodies are heterodimeric protein structures composed of a light chain and a heavy chain. There are five different human Ig isotypes (IgM, IgG, IgA, IgE, and IgD), which differ in the structure of the constant region. The antigen-binding site is composed of the variable regions of the heavy and light chains. Both chains contain a constant region and a variable region that undergoes recombination and somatic hypermutation. The variable (V), diversity (D), and joining (J) gene segments of the Ig heavy and light chain loci rearrange through DNA recombination to produce a diversified B-cell repertoire. However, BCRs with higher affinity confer enhanced capacity to interact with Tfh cells, therefore favoring the selection and expansion of the most effective B-cell clones.

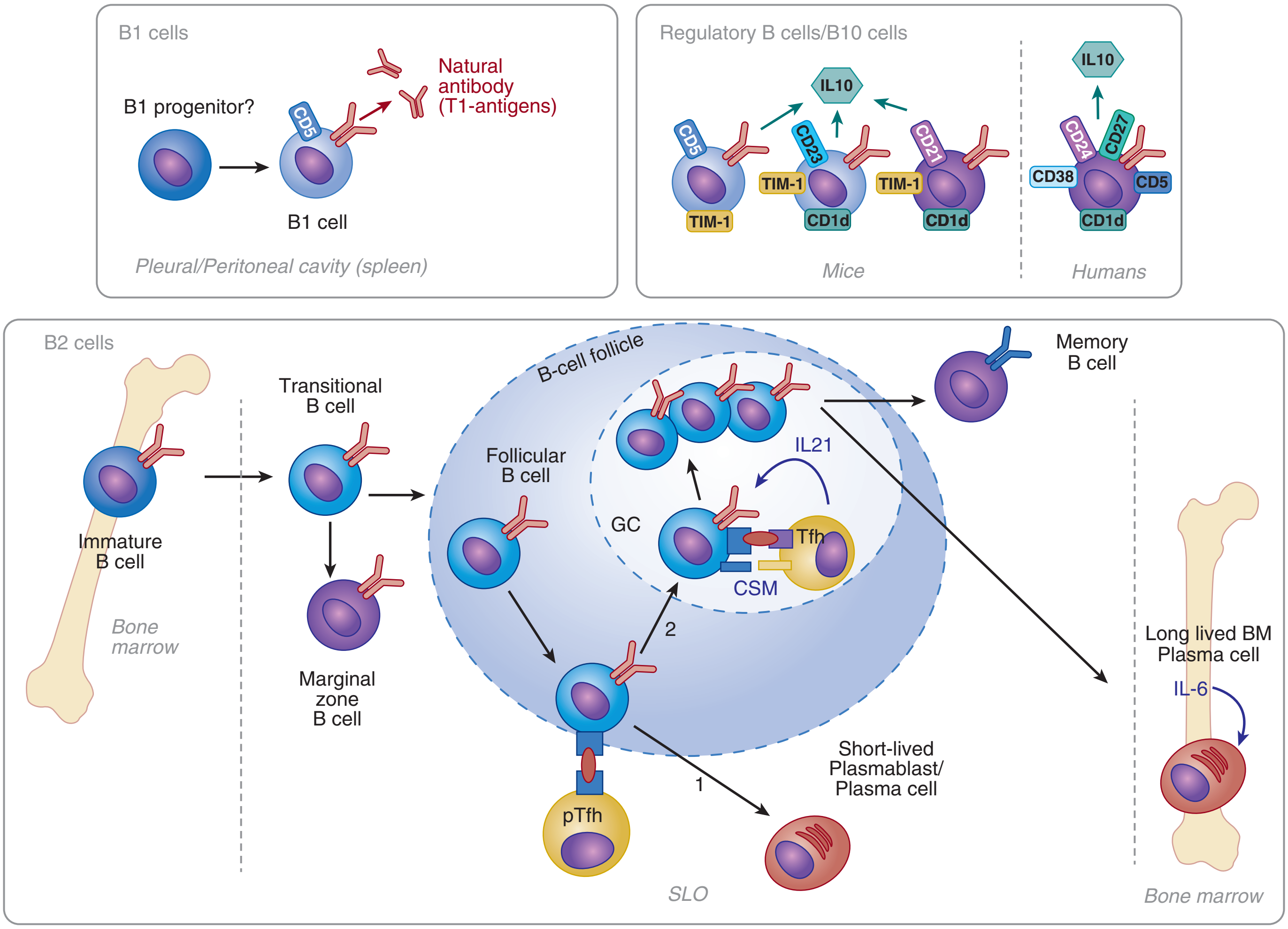

Fig. 69.5 B-cell subsets. B1 cells reside within pleural and peritoneal cavities (and in smaller numbers in the spleen). They produce low-affinity natural antibody to T-independent antigens. B2 cells are formed in the bone marrow and develop from pro-B cells, to pre-B cells, through to immature B cells, which are released into the periphery. Following the antigen encounter, follicular B cells move to the T–B border within secondary lymphoid organs (SLO) and present antigen to a cognate CD4 T cell, known as a pre-T follicular helper cell (pTfh). Following this interaction, the fate of the B cell lies between two pathways: 1) To become a short-lived extrafollicular plasmablast or 2) to enter the germinal center, undergo rounds of somatic hypermutation and selection to generate class-switched memory B cells or plasma cells. This process requires the presence of IL-21-producing T follicular helper cells (Tfh). The interactions between Tfh and B cells also requires costimulatory molecule (CSM) interactions, including CD40L/CD40, ICOS/ICOS-L, and CD28/CD86. Some B cells have the capacity to produce the immunoregulatory cytokine IL10 and have been dubbed *regulatory B cells* or *B10 cells*. (From Clatworthy MR. B-cell regulation and its application to transplantation. *Transpl Int*. 2014;27(2):117–128. With permission.)

While long-lived plasma cells home to the bone marrow to continuously produce high-affinity antibodies, memory B cells require antigen rechallenge to initiate antibody production.

Mounting evidence suggests that B-cell activation, maturation, and antibody production can also occur within tertiary lymphoid structures developing within allografts. These structures are found in association with chronic inflammatory conditions, including autoimmune disorders, infection, cancer, and transplantation. Tertiary lymphoid structures are reminiscent of germinal centers, with compartmentalization of T and B cells and the presence of DCs. They have been reported in animal models of rejection and in chronically rejected human renal and cardiac allografts. Mounting evidence suggests that anti-HLA antibodies and autoantibodies deleterious to the allograft are produced within tertiary lymphoid structures.[39–41] The production of lymphotoxin-β by B cells can favor the formation of tertiary lymphoid structures.

The current classification of rejection establishes a distinction between T-cell–mediated and antibody-mediated rejection episodes. However useful for histopathologic classification, it is important to remember that in most cases, the activation of naïve B cells leading to the production of donor-specific antibody (DSA) requires T-cell help. However, if the production of long-lived plasmablasts or memory B cells is established prior to transplantation because of allosensitizing events, such as pregnancy or prior transplantation, the need for T-cell help is bypassed. Autoantibodies have also been associated with either increased risk of acute rejection or reduced long-term allograft survival,[42] as will be discussed in detail later in this chapter. The role of T-cell help in the production of these autoantibodies remains unclear.

ANTIHUMAN LEUKOCYTE ANTIGEN ANTIBODIES AND GRAFT INJURY

DSA can injure the allograft through various pathways, including complement activation, antibody-dependent cell-mediated cytotoxicity (ADCC), and induction of endothelial dysfunction.

COMPLEMENT ACTIVATION

Complement is an important component of the innate immune response and plays a major role in clearing damaged cells and invading pathogens.[43,44] The complement pathway is composed of cell surface receptors, regulatory proteins, and soluble proteins that can interact and cleave one another to activate downstream components. The complement pathway consists of three distinct activating components: Classical, lectin, and alternate pathways (Fig. 69.6).[43] All three pathways converge into a common effector phase initiated by the formation of a C3 convertase, which results in the formation of the membrane attack complex C5b-9 that precipitates cell lysis. The classical pathway is activated when C1q binds to IgG or IgM antibodies present within immune complexes. The general order of complement fixing activity for antibodies is IgM>IgG3>IgG1>IgG2≫IgG4. IgE antibodies are non-complement fixing, whereas IgA can activate the alternative, but not classical pathway. The binding of C1q to complement-fixing antibodies leads to interactions with the serine proteases C1r and C1s, allowing for the cleavage of C4 and C2, and the formation of the multiprotein complex and C3 convertase, C4b2a. The classical pathway plays an important role in antibody-mediated rejection. C4d staining, as used in the diagnosis of antibody-mediated rejection, is a read-out of C4 cleavage, and is considered a good marker of classical pathway activation. The lectin pathway is activated by the interaction of mannose-binding lectin (MBL), ficolins, and collectin-11 with carbohydrate pathogen-associated molecular patterns (PAMPs) and DAMPs, which are expressed on the surface of pathogens or damaged cells. Like the classical pathway, the lectin pathway leads to the formation of the C4b2a C3 convertase and is therefore an additional source of positive C4d staining. The lectin pathway is thought to play an important role in ischemia-reperfusion injury.[43] Lastly, the alternative pathway is activated in circumstances where soluble inhibitors cannot keep basal C3 hydrolysis in check, therefore allowing for interactions between C3, factor B and factor D. This leads to the formation of the C3 convertase C3bBb. Increased activation of the alternative pathway occurs in the presence of various factors associated with tissue injury, including microparticles, platelet surfaces, and endotoxins. C3 convertases (either C4b2a or C3bBb) cleave C3 into C3a, which has proinflammatory and chemotactic functions, and C3b, which in turn cleaves C5. This leads to the formation of C5a, another potent chemotactic and proinflammatory mediator, and C5b. The latter interacts with distal complement components C6 to C9, leading to the formation of a lytic complex, C5b-9, that subsequently creates a pore in the plasma membrane of the target cell.

A number of soluble regulatory molecules can keep complement activation in check. Factor H accelerates degradation of key components of the alternative pathway and serves as a cofactor for factor I. The latter accelerates degradation of C3b and C4b. C1 esterase inhibitor and C4b binding protein interfere with the activation of the classical pathway. Cell membrane regulatory proteins include decay-accelerating factor (DAF, CD55), membrane cofactor protein (CD46), and CR1 (CD35). During antibody-mediated rejection, DSA and autoantibodies interact with antigens present on the microvascular endothelium, leading to activation of the classical complement pathway and lytic injury and inflammation of the microvasculature.

ANTIBODY-DEPENDENT CYTOTOXICITY

Interactions between antibodies and cognate antigens, such as MHC molecules or autoantigens expressed on the allograft endothelium, can also favor the recruitment and activation of innate immune effectors that express Fc receptors, such as monocytes, macrophages, DCs, and NK cells.[45] There are three classes of Fc receptors, FcγRI (CD64), FcγRII (CD32), and FcγRIII (CD16), the latter two of which both have A and B isoforms. All Fc receptors, except FcγRIIIB, are activating. NK cells and innate types of lymphoid cells are considered central effectors of ADCC. They express FcγRIIIA in the absence of inhibitory FcγRIIIB.[46,47] Binding of antibodies to Fc receptors activates NK cells, leading to the release of cytotoxic granules such as perforin/granzyme B. Indeed, renal biopsies from transplant patients with DSA show higher NK cell transcripts and increased numbers of NK cells within peritubular capillaries.[48]

ENDOTHELIAL ACTIVATION

Antibodies that target endothelial cells can also injure renal allografts by inducing endothelial activation and dysfunction.

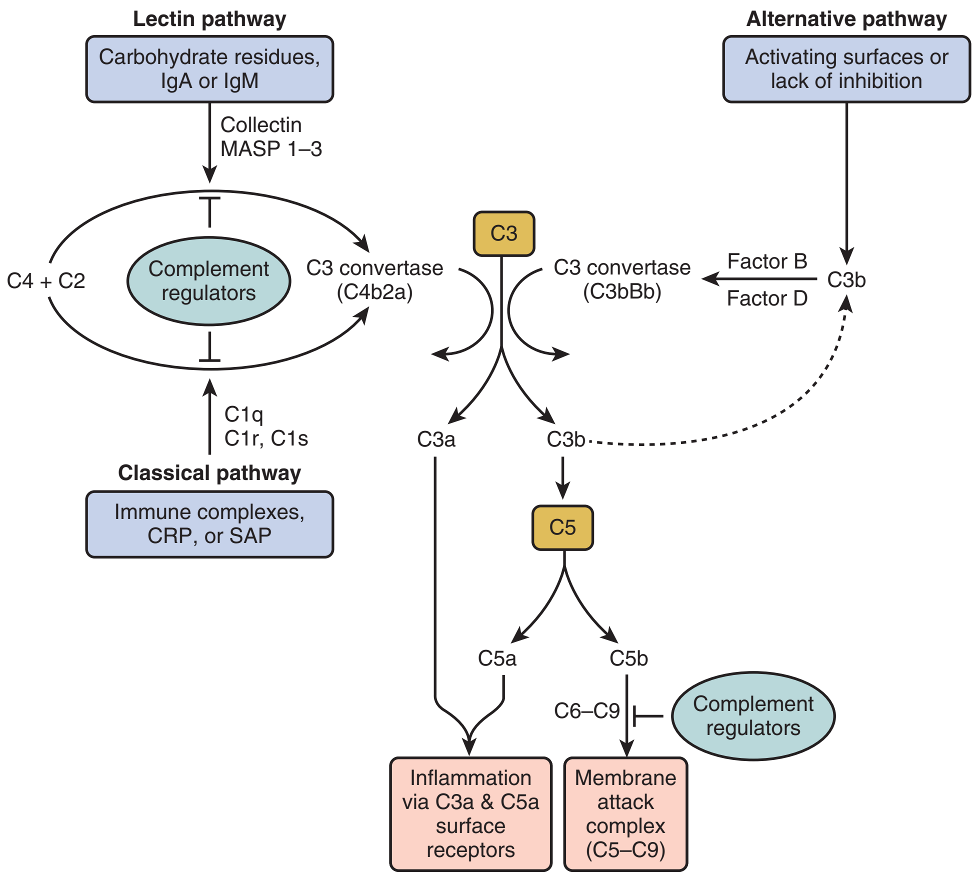

Fig. 69.6 The complement cascade. The complement system is activated by one of three major pathways: classical, lectin, or alternative. The classical pathway is triggered by C1 binding to immune surveillance molecules, such as IgG, IgM, C-reactive protein (CRP), or serum amyloid protein (SAP), which are attached to the target sequence. The LP is triggered by the binding of collectins, such as MBL and collectin-11, or ficolins to carbohydrate residues on a pathogenic surface or IgA and IgM molecules. The alternative pathway is initiated by direct binding of C3b to activating surfaces. All three pathways converge at the production of the central complement component C3. That is, all pathways form enzyme complexes (classical or alternative convertases) that cleave either C3 (into C3a and C3b) or C5 (into C5a and C5b). C5b triggers the terminal pathway by creating a pore in the target cell membrane via the formation of the membrane attack complex (C5b-C9). Soluble complement effectors C3a and C5a are detected by specific cell receptors thereby promoting inflammation. Complement inhibition occurs via a variety of molecules ultimately inhibiting C3 and C5 convertase or blocking the formation of the membrane attack complex (C5b–C9). (From Sacks S et al. Complement recognition pathways in renal transplantation *J Am Soc Nephrol.* 2017; 28: 2571–2578, Fig 1).

Anti-HLA antibodies cross-linking cognate HLA molecules at the surface of endothelial and vascular smooth muscle cells can initiate signal transduction pathways that favor cytoskeletal rearrangement, proliferation, activation, exocytosis, and recruitment of leukocytes. Activation of the SRC/FAK/PI3K/AKT pathway by integrin signaling, as triggered by anti-HLA antibodies, activates the mTOR signaling pathway, favoring proliferation and accumulation of vascular cells. Collectively, these changes are thought to favor neointimal thickening and mononuclear leukocyte infiltration.[49]

OTHER TRANSPLANT-RELEVANT ANTIGENS AND ANTIBODIES

DSA have classically been involved in the majority of cases of acute and chronic antibody-mediated rejection (ABMR). However, ABMR has also been documented in patients who do not have circulating DSA.[50,51] Although DSA can be adsorbed within the allograft,[52] mounting evidence has pointed to the role of non-HLA antibodies as important contributors to ABMR. The most recent versions of the Banff classification system now recognize that, in addition to the presence of DSA, antibodies targeting other donor antigens can be considered as one of the diagnostic criteria for acute or chronic ABMR.[53,54]

The MHC I-related chain gene A (MICA) is located within the MHC I region of chromosome 6.[55] The MICA protein is highly polymorphic, has an extracellular domain, is constitutively expressed on endothelial cells, and its expression has been documented in kidney graft biopsies.[56–58] Taken together, these features make MICA an antigenic target of potential clinical relevance in solid organ transplantation. The demonstration that MICA was indeed a target for complement-dependent cytotoxicity[59] was followed by clinical studies showing that pretransplant MICA antibodies were associated with ABMR[60–62] and decreased graft survival.[61] However, two recent studies failed to show an association between pretransplant MICA status and ABMR or graft survival,[63,64] which may relate to differences in the strength of the immunosuppressive regimen used.[55] The detection of anti-MICA antibodies posttransplant has also been linked to an increased risk of ABMR,[65,66] chronic rejection,[67] and decreased graft survival[68,69] in some, but not all studies.[70] In addition to studies yielding discordant results, whether the anti-MICA antibodies identified are specific to the donor has rarely been assessed.[55] Hence the association between anti-MICA antibodies and graft outcomes remains controversial.

The proximity of donor endothelial cells to the recipient's immune system makes endothelial antigens likely targets for antibody-mediated injury. Anti-endothelial cell antibodies (AECA) have been detected in prekidney transplant patients, regardless of whether they are HLA-sensitized or desensitized.[71] Hyperacute rejections involving AECA but not anti-HLA DSA have been described, suggesting that these antibodies can be pathogenic.[72,73] The appearance of de novo, posttransplant AECA has been linked to adverse graft outcomes,[74,75] although the relevance of pretransplant AECA on graft outcomes is controversial.[61,72,74,76] These discrepancies may stem from the variability in the methods used to measure AECA and the lack of standardized assays,[77] given that the exact antigenic targets had not been uncovered when these studies were conducted. Recent progress on this issue stems from the identification of potential targets for AECA. Using a proteomic approach, novel endothelial antigenic targets (endoglin, Fms-like tyrosine kinase-3 ligand, EGF-like repeats, discoidin I-like domains 3, and intercellular adhesion molecule 4) have recently been characterized from antibody eluates in kidney transplant patients with DSA-negative ABMR. Reactivity of pretransplant serum to these antigens was associated with an increased risk of ABMR in a validation cohort.[78] Keratin 1 has also been identified as a potential antigenic target for AECA.[79]

In addition to alloantibodies, emerging evidence shows that autoantibodies are actively involved in acute and chronic rejection in kidney,[80–84] heart, and lung transplant recipients.[85,86] For instance, polyreactive, natural IgG autoantibodies against apoptotic Jurkat cells that can activate complement were isolated from the sera of kidney transplant recipients with ABMR.[84] In a retrospective cohort study, the presence of these antibodies before transplantation was associated with decreased graft survival.[87] Agonistic autoantibodies against angiotensin II receptor type 1 (AT_1R-Abs) have been associated with acute vascular rejection in DSA-negative renal transplant patients.[80] The passive transfer of these autoantibodies in an animal model reproduced the phenotype observed in transplant patients, demonstrating the pathogenic impact of these autoantibodies.[80] Multiple studies have shown that the presence of pretransplant AT_1R-Abs is associated with an increased risk of rejection and graft loss.[88–90] A recent study has questioned the association between AT_1R-Abs, rejection, and graft failure in the era of Luminex-based DSA screening.[91]

Autoantibodies against components of the vascular basement membrane such as the LG3 fragment of perlecan or agrin[42,81] have been associated with acute vascular rejection or transplant glomerulopathy.[83] Similarly, de novo and pretransplant levels of antifibronectin and anticollagen type 4 autoantibodies were linked to transplant glomerulopathy in kidney transplant patients.[82]

INTERACTIONS BETWEEN ISCHEMIA-REPERFUSION AND AUTOANTIBODIES

Ischemia-reperfusion, a key feature of the solid organ transplantation process, seems to enhance both the impact and the production of autoantibodies. In contrast to HLA and MICA, which are constitutively expressed on the cell surface of the allograft endothelium, autoantigens are usually cryptic and only become exposed following ischemia-reperfusion-induced tissue damage.[42,92] In line with this concept, ischemia-reperfusion was found to enhance the contractile effect of AT_1R-Abs on isolated aortic rings and the impact of anti-LG3 on vascular rejection in an animal model of aortic transplantation.[81,93] In mice, passive transfer of anti-K-α-1 tubulin, an autoantibody associated with chronic lung rejection, increased airway inflammation in an animal model of autologous lung transplantation.[94] Taken together, these studies suggest that ischemia-reperfusion can increase the availability or affinity of antigenic targets, as well as enhance the deleterious impact of autoantibodies on the transplanted allograft.[42,81,93,94] Recent data also suggest that ischemia-reperfusion can fuel the production of autoantibodies in the context of transplantation. In a cohort of stable pediatric kidney transplant recipients, an enrichment in autoantibodies directed at antigenic targets from the renal pelvis and medulla, areas that are particularly sensitive to ischemia, was observed when pre- and posttransplant sera were compared.[95]

INTERACTIONS BETWEEN ALLOANTIBODIES AND AUTOANTIBODIES

In addition to their interaction with ischemia-reperfusion, some transplant-relevant autoantibodies seem to promote the development of alloantibodies. In a cohort of kidney transplant patients, the presence of pretransplant AT_1R-Abs was an independent risk factor for the development of *de novo* DSA.[96] In lung transplant recipients, pretransplant positivity for autoantibodies against lung-restricted antigens, such as K-α-1 tubulin, was associated with the production of *de novo* DSA and chronic lung rejection.[86] This association could be explained by both autoantibody-mediated endothelial activation, which increases the expression of HLA targets on the graft endothelium, and by the recipient's overall propensity to mount an antibody response. Autoantibodies can also enhance the effect of alloantibodies. Indeed, kidney transplant patients with abnormal kidney biopsies who were positive for both AT_1R-Abs and DSA experienced a higher risk of graft loss compared with those who were only DSA positive.[97]

MECHANISTIC EFFECT OF NON–HUMAN LEUKOCYTE ANTIGEN ANTIBODIES

Although mechanistic pathways for non-HLA antibodies have not been studied to the same extent as anti-HLA antibodies, they may also cause graft injury through the activation of endothelial cells or complement, and/or the promotion of leukocyte interaction/activation.[98] AECA activate endothelial cells, increase the expression of HLA class 1 and adhesion molecules (E selectin, ICAM1), and promote the production of inflammatory cytokines such as platelet-derived growth factor (PDGF).[78] AT_1R-Abs mediate endothelial cell activation and vasoconstriction by binding to the second extracellular loop of the angiotensin II receptor type 1 protein, therefore promoting the activation of NF-κB and downstream cytokines.[80] Anti-MICA, anti-LG3, and polyreactive antibodies to apoptotic cells mediate injury, at least in part, through complement activation.[42,59,81,84,99]

HISTOCOMPABILITY TESTING FOR THE KIDNEY TRANSPLANT CANDIDATE AND RECIPIENT

In the previous section, we discussed mechanisms of allorecognition and effectors pathways inducing allograft rejection. Because HLA are the most important foreign antigens recognized on kidney allografts, this section will cover this topic in greater depth, focusing primarily on clinical translation of HLA basic science into clinical testing considerations. The Histocompatibility Laboratory (HLA Laboratory) can be considered as the site of applied transplant immunobiology. The HLA Laboratory offers clinical immunologic risk consultation, with the purpose of estimating immunologic risk throughout transplant assessment, consideration of potential donor(s), and the posttransplant period. Indeed, since the first clinical histocompatibility testing was defined in 1969,[100] the laboratory has grown to play a critical role in transplant decision making and allocation science. Histocompatibility testing (henceforth called HLA testing) focuses predominantly on assessing the humoral branch of the alloimmune response and does not routinely assess directly for activity within or risk associated with the cellular compartment of the immune response. Notwithstanding the clear interplay between cellular and humoral alloreactivity, there is a clear and strong association with measurement of both histocompatibility and the humoral response and clinical outcomes (ABMR, transplant glomerulopathy and premature graft loss). Hence precise and thorough delineation of the presence and risk of memory and de novo humoral alloreactivity to non-self HLA antigens is required. This section will focus on the most common platforms of testing utilized in the modern HLA laboratory, their strengths and limitations. We will describe how these platforms can support humoral immunologic risk assessment that can be translated into clinical action with the goals of increasing transplant access, and reducing allograft damage related to immunologic incompatibility, optimizing both equity and utility. As discussed in the previous section, antibodies to non-HLA antigens may also be correlated with worse allograft outcomes or augment the risk portended by HLA antibodies. At present, routine screening for these non-HLA antibodies is not common in clinical programs but may occur on a case by case basis.

Three testing platforms have formed the core of clinical transplant immunologic testing for more than 40 years: HLA typing (of alleles with determination of their corresponding antigen(s)); HLA antibody screening and identification, and donor-specific crossmatching. Clearly, there have been significant advances in the methods themselves but the foundational principles of using their results in context to quantify donor and recipient HLA antigen differences, as well as HLA antibody reactivity to potential donor(s), remain unchanged. We will focus on the most current methodologies here, with mention of historical, largely obsolete methods, only briefly for reference in the context of interpreting older literature based on these tests. This section will concentrate on the three core HLA testing methods, their interpretation considerations, their limitations and their practical utilization in clinical transplantation.

HUMAN LEUKOCYTE ANTIGEN TYPING

CONTEXT

Outside of transplantation, HLA antigens are themselves the antigen presenting molecules, alerting the immune system to foreign peptides (for example, infectious or malignancy derived peptides). Clearly significant diversity in the HLA repertoire is beneficial for ensuring effective vigilance against environmental threats within and between populations. However, it is this diversity that gives the HLA antigen its potent immunogenicity in transplantation, cognizant that the immune system cannot distinguish the threat potential of an infectious agent from that associated with a foreign HLA antigen on a transplanted organ. Indeed, there are more than 10,000 unique HLA alleles coding for more than 8000 distinct HLA proteins reported across the human repertoire, where each person has up to 18 unique HLA proteins on the surface of their nucleated cells. In sibling relationships HLA identity can occur by chance in 25% of pairs through Mendelian inheritance of the short arm of chromosome 6 where the HLA proteins are coded, but outside of this relationship, complete HLA identity between randomly selected donors and recipients is indeed rare. Hence, in most cases, the goal of HLA typing is not to find HLA identical donor–recipient pairs but to quantify the degree of difference between donors and recipients (as one estimate of the stimulus of the alloimmune response) and to determine if a recipient carries HLA antibodies specific to a given donor, driving clinical decision making accordingly. Indeed, any degree of mismatch between a donor and a recipient can stimulate an alloimmune response in the absence of enough immunosuppression, leading to subclinical and clinical rejection.[101–109]

HUMAN LEUKOCYTE ANTIGENS

HLA class I molecules (HLA-A, B, C) are found on the surface of all nucleated cells, whereas the expression of HLA class II molecules (HLA-DR, DRw, DQ, DP) is limited to B cells, APCs, and endothelial cells. The expression of class II molecules on endothelial cells is particularly important in the transplanted organ[109–111] that may undergo injury during or after transplantation, upregulating these very important immune targets. Class I molecules are comprised of a polymorphic chain encoded by the HLA class I gene, supported by a monomorphic β_2-microglobulin. As such, the immunogenicity of the class I molecule is defined by the polymorphism in this single chain. Conversely, although the three-dimensional structure of class II antigens is similar to class I (Fig. 69.7), their amino acid configuration is different; the molecule is comprised of two distinct chains (alpha and beta), each with inherent polymorphism. The molecule's immunogenic potential in transplantation is therefore present within each amino acid chain, but also with additional immunogenicity potential at the interface of the alpha and beta chains. It

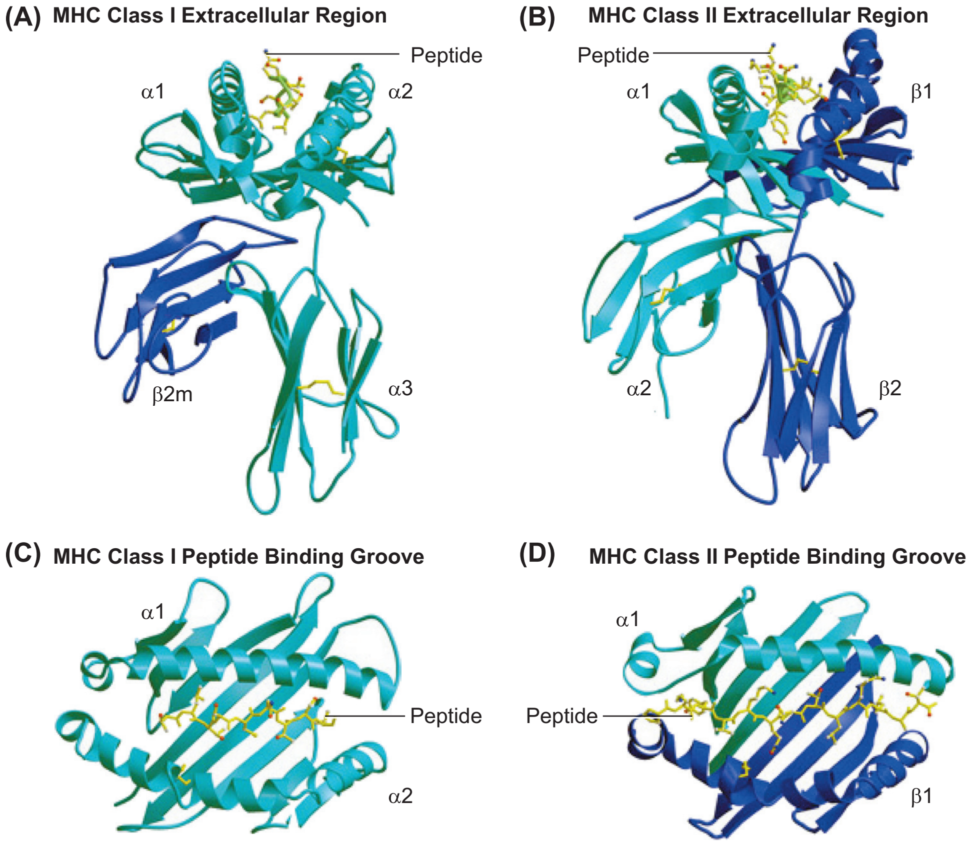

Fig. 69.7 X-ray crystal structures of MHC Class I and II molecules in the mouse crystal structures showing the carbon backbone of murine MHC class I and II molecules. (A) and (B) show peptides bound to the extracellular regions of MHC class I or II, respectively. (C) and (D) show the view looking down at the peptide in the peptide-binding groove of MHC class I or II, respectively. (Reproduced by permission of Bjorkman PJ. MHC restriction in three dimensions: A view of T cell receptor/ligand interactions. *Cell.* 1977;9:167–170.)

is in this complex and polymorphic structure of the HLA molecules that the immunologic challenges of transplantation begin: Even single amino acid differences between a donor and recipient HLA, if sufficiently different in charge, size or polarity, can trigger alloimmune pathways in the recipient.

HUMAN LEUKOCYTE ANTIGEN TYPING METHODS

SEROLOGIC METHODS

Historically, HLA antigens were identified using common serologic methods. Sera containing HLA antibodies of both single and multiple specificities to broad groups of HLA antigens were obtained from sensitized patients (usually multiparous women) and specifically chosen in combination in panels to cover the most common HLA antigen types in a population. The lymphocytes of the individual to be typed were then mixed with all serum samples in the panel in the presence of complement and a vital dye. If the sera in each reaction resulted in cell death with complement visualized by vital dye uptake, it could be inferred that the antibody(ies) in that serum had specificity(ies) to one or more HLA antigens on the cell surface. Careful examination of the pattern of positive reactions across the test sera would yield the predicted HLA antigens on the lymphocytes being tested.

MOLECULAR METHODS

Several molecular methods have been developed to better characterize HLA antigens by determining more precisely the HLA alleles that encode them, and in doing so, more accurately identify the amino acid differences between different HLA antigens (and indeed even between different alleles within the same antigen group). Currently utilized methods use polymerase chain reaction (PCR) platforms, and options include the real time PCR (rtPCR), sequence-specific primer (SSP) method, the reverse sequence-specific oligonucleotide (R-SSO) probe method, Sanger-based sequencing, and, most recently, next-generation sequencing. It warrants mention that the sequencing methods are not suitable for deceased donor typing, as they can take several days to complete; these are more commonly employed for recipient typing and living donor typing when used. Conversely, rtPCR, SSP and SSO methods are more commonly used for deceased donor typing because their rapid turnaround times facilitate prompt transplant decision making in allocation. The details of each method will not be reviewed here, as methods are in rapid evolution and chosen in the context of what is appropriate for a given program's clinical needs. Readers are directed to their own HLA laboratory for a more complete review of the local methods used. Regardless of the specific molecular platform employed, molecular methods as a group allow for a more precise determination of HLA alleles and their corresponding immunogenicity, where serologic typing would fail to identify many of these small but important differences.

INTERPRETING HUMAN LEUKOCYTE ANTIGENS TYPING RESULTS

NOMENCLATURE—ALLELES AND ANTIGENS

Molecular methodology for HLA typing uses a precise nomenclature to identify HLA alleles; it uses a letter to identify the locus (A, B, C, DRB1, DRB3/4/5, DQA1, DQB1, DPA1, DPB1), an asterisk, which confirms that molecular methods were used, and then a series of numbers separated by colons (called field separators) to identify precisely the unique allele.[112]

For example, HLA-A*03:01:01:02N refers to a molecular defined allele at the A locus, followed by four unique fields, each of which has a precise meaning. Field 1 (in this example 03) identifies the allele group. Field 2 (in this example 01) identifies a precise allele, with at least 1 amino acid difference in the mature protein that distinguishes it from all other alleles in the allele group. Field 3 identifies if the genotype of the allele contains a synonymous DNA substitution in a coding region that does not alter the final protein structure, and field 4 indicates a difference in DNA sequence in a noncoding region of the gene. Additionally, a suffix may be appended to the allele name (in this example N) indicating a variant of expression of the final protein product, where N represents Null or No expression; L indicates Low expression; S indicates a molecule that is Secreted and Soluble but not present on cell surfaces; C is assigned to alleles producing products only detected in the Cytoplasm; A indicates Aberrant expression, where the expression is not confirmed; and Q is assigned to Questionable, where the impact of the mutation on expression is not confirmed but has been seen in other alleles.

For practical purposes in solid organ transplantation, only the first two fields have clinical impact in so far as they determine the protein to be considered in mismatching and antibody production. Additionally, the N or L suffix is important in transplantation, as where a protein has null or low expression it may not have any (null) or reduced (low) relevance in immunogenicity or antibody specificity considerations. Indeed, most commonly at the present time molecular typing in kidney transplantation is reported at only the first field level unless the specific allele is needed to quantify differences between a donor and recipient or to interpret a specific antibody pattern as donor specific or not.

Molecular typing nomenclature also permits a more accurate description of the complexity of the class II DQ and DP proteins, where the antigen is named only for the beta chain, but the immunologic relevance of the molecule requires identification of the alpha chain distinctly and in combination with the beta chain[113–117] (Fig. 69.7). Indeed, the authors recommend to always identify DQ and DP antigens by both their alpha and beta chains to properly describe the encoded final protein and any antigen–antibody interactions involving the alpha chain. The DRA1* chain is not sufficiently polymorphic such that naming the DR antigens in accordance to their beta chain allele is immunologically enough.

Table 69.1 Common Human Leukocyte Antigen Genes With Multiple Human Leukocyte Antigens Associated

Human Leukocyte Antigen Gene	Antigens Encoded by Gene
HLA-B*14	B64 or B65
HLA-B*15	B62, B63, B71, B72, B75, B76, or B77
HLA-B*40	B60 or B61
HLA-C*03	Cw9 or Cw10
HLA-DRB1*03	DR17 or DR18
HLA-DQB1*03	DQ, DQ8, or DQ9

By way of comparison, HLA antigens (serologically determined) are identified by the locus (with no asterisk) and a single number that indicates the specific protein being expressed; for example, A2 or B27. This format specifically indicates a protein rather than an allele and, in fact, represents the group of all A2 proteins; that is, the protein product of all the HLA-A*02 alleles. This nomenclature is particularly important because whenever HLA antibodies are identified (see later), the antibody is most typically named for its target antigen using antigen nomenclature. Generally, the gene group from the molecular name and the antigen number are both familiar and similar; for example, HLA-A*02 gene group codes for A2 proteins and HLA-B*27 group codes for B27 proteins. However, there are some exceptions noted (Table 69.1). Typically, in these cases the molecular typing is followed by the relevant antigen in parenthesis; for example, HLA-DQB1*03(7), where the DQB1*03 gene codes for DQ7 protein.

MATCHING AND MISMATCHING OF ALLELES, ANTIGENS AND EPITOPES

Donors and recipients may be compared by identifying the differences between their HLA antigens and alleles; however, directionality is critical in describing this accurately. Biologically, the most relevant comparison in transplantation is to describe the donor differences to the recipient from the perspective of the recipient's immune system, and to describe the donor antigen or allele differences in terms of the number of mismatches in the host versus graft (HvG) direction. This is particularly important when either the donor or recipient is homozygous for one or more HLA antigens or alleles (Table 69.2). Newer analytic methods such as HLA Matchmaker[118–120] also allow laboratories now not only to compare allele and antigen differences from a donor to a recipient but also to identify and count the number of eplet (3–4 amino acid differences at positions with immunogenicity potential) and epitope (structure surrounding the eplet where the Fab' can bind) differences, typically expressed as the total number of eplet differences or eplet "load." Methods such as these can also identify specific eplets that are different between donor–recipient pairs that may have greater immunologic relevance.[121–126]

LIMITATIONS OF HUMAN LEUKOCYTE ANTIGEN TYPING METHODS

Serologic methods identify only the broadest of HLA antigens at their most common and strongest humoral targets;

Table 69.2 Human Leukocyte Antigens Mismatching Considered From the Perspective of the Recipient Immune System

Human Leukocyte Antigen Locus	A*	B*	Cw*	DRB1*	DQA1*	DQB1*	DPA1*	DPB1*	Total Mismatches
Recipient	11	15(62) 15(75)	04	12	06:01	03:01(7)	01:03	02:02	
	24		08	14	–	–	02	21:01	
Donor	2	46	01	09	03	03:03(9)	02-	05:01	
	–	52	14	–	–	–	–	09:01	
Mismatch	1	2	2	1	1	1	0	2	10
Recipient	2	46	01	09	03	03:03(9)	02-	05:01	
	–	52	14	–	–	–	–	09:01	
Donor	11	15(62) 15(75)	04	12	06:01	03:01(7)	01:03	02:02	
	24		08	14	–	–	02	21:01	
Mismatch	2	2	2	2	1	1	1	2	13

Molecularly determined alleles at each human leukocyte antigen (HLA) locus in recipient and donor in two different scenarios (upper and lower section of table), showing the number of mismatches.

In the case where either a recipient or donor is homozygous for one or more antigens, there is a difference in mismatches depending on which direction the mismatches are considered. In the first example, considering the foreign HLA donor, antigen burden yields 10 mismatches in the host versus graft direction. However, if the donor and recipient had the opposing HLA typing, it would give a total of 13 mismatches from the perspective of this recipient.

Clinical Relevance

The authors encourage the reader to interpret any historical data limited to HLA-A, B, and DR typing with caution considering modern-era data supporting the immunogenic role of proteins from HLA loci in transplant outcomes. When interpreting molecular results, the clinician should be sure to note whether allele level typing (second field resolution) is in fact necessary to interpret immunologic risk either at the eplet mismatch level or antibody–antigen interaction level. Consultation with histocompatibility specialists should be sought to determine the level of resolution needed in any given case.

they do not distinguish, in most cases, between small but immunologically important differences that may be present at alleles within the same antigen group. This is particularly relevant when HLA antibodies (see later) have only a subset of allele-coded proteins within a group (allele-specific antibodies) as their target. Additionally, serologic methods only commonly identified with any accuracy A, B and DR proteins, giving rise to the (now recognized as incorrect) paradigm that these six antigens were the most significant in transplantation. The authors encourage the reader to interpret any historical data limited to HLA-A, B, DR typing with caution, considering modern era data supporting the immunogenic role of proteins from HLA loci in transplant outcomes. Molecular methods in general have advanced the understanding of HLA diversity and its contribution to alloimmunity, but different platforms have different levels of typing resolution and all are under rapid evolution such that any specific discussion of limitation would shortly be obsolete. At present, the major limitation to some methods that result in a high level of resolution is that several days may be required to obtain a result, which is prohibitive in deceased donor transplantation workup and allocation; this, however, is anticipated to change with the rapid improvement of methods and platforms. Rather, when interpreting molecular results, the clinician should be sure to note whether allele level typing (second field resolution) is in fact necessary to interpret immunologic risk, either at the eplet mismatch level or antibody-antigen interaction level. Consultation with Histocompatibility specialists should be sought out to determine the level of resolution needed in any given case.

USING HUMAN LEUKOCYTE ANTIGEN TYPING RESULTS IN KIDNEY TRANSPLANTATION

INTERPRETING THE TRANSPLANT CANDIDATE'S ANTIBODY PROFILE

The first step in evaluating a transplant candidate is to determine their own HLA allele and antigen profile using the highest resolution typing available in that center. This information is useful immediately, well before a donor is under consideration, as the recipient's HLA antibody profile must always be interpreted in the context of their own HLA typing to ensure that the antibodies identified are as accurate as possible (cognizant that one does not make HLA antibody to self-HLA antigens or self-epitopes). We note that antibodies can form to self-eplets,[126] but only when the surrounding epitope is non-self. These considerations are crucial for accurate antibody interpretation, and sufficient resolution of recipient HLA typing is thus of utmost importance. Furthermore, in very highly sensitized patients, understanding the population frequency of the candidate's own HLA alleles and haplotype can aid in the estimate of donor access (likelihood to receive a donor organ), with those patients who have rarer HLA alleles potentially having an even further reduced access compared with those with more common HLA phenotypes.

CLINICAL ESTIMATE OF IMMUNOGENICITY OF HUMAN LEUKOCYTE ANTIGEN MISMATCHES

The number of HLA mismatches between donor and recipient was one of the first recognized predictors of transplant outcome as early as 3 years posttransplant, even in the early days of transplantation, when typing was only resolved at the serologic level and assessment was limited to only a few loci.[127–130] With improved immunosuppression, the impact of mismatching at the antigen level was reduced, and its role in allocation correspondingly diminished.[131,132] Studies using larger data sets and performed in the era of modern immunosuppression, with a view to longer-term clinically relevant outcomes, however, still show that donor–recipient HLA mismatching, even at the antigen level, can have a deleterious impact on graft survival.[133–136] Notably, however, all HLA antigens are not equally different at the eplet/epitope level, and simply considering the integer difference in mismatches is at best a crude estimate of immunogenicity (Table 69.3). Eplet analysis is a more granular and biologically relevant estimate of immune differences at the level of the HLA molecule. Early data support this strategy with HLA eplet load being strongly associated with both *de novo* DSA development[123,137] and clinically relevant outcomes, such as transplant glomerulopathy.[124] It is further recognized that not all HLA eplets are immunologically equivalent. As such, eplet load is itself also only an estimate of immune difference. There is much interest in determining which eplet differences carry the greatest risk of adverse outcome, but such data are currently not yet available.

As mentioned earlier, the role of HLA matching in allocation has been reduced in many jurisdictions. This is both because of the reduced impact on short- and medium-term outcomes, as well as concerns about creating biases against individuals of certain races/HLA phenotypes, where their group's HLA antigens are less commonly found among typical organ donors. There is insufficient evidence available at this time to allocate organs strictly based on eplet load or eplet matching. However, this may evolve with rapidly emerging data. How then should the clinician interpret HLA mismatch data at the antigen, allele, or eplet level? It is important to acknowledge that if the mismatches in a given pair are greater, then the immunologic potential for alloimmune recognition and response is also greater. The clinician should use this information to refine the risk assessment along a continuum. Higher-risk states may warrant greater vigilance in follow-up (lab tests, HLA antibody monitoring) or greater caution in reducing immunotherapy.

AVOIDANCE OF UNACCEPTABLE DONOR ANTIGENS/ALLELES

Foundational to modern organ allocation is evaluating donor HLA typing in the context of a recipient's HLA to identify if there are any DSA. This is commonly referred to as virtual crossmatching (VXM), where positive DSA is interpreted as a positive VXM and the absence of DSA is a negative VXM. This is discussed in more detail in the section on virtual crossmatching later. From the perspective of HLA typing, it is critical that donor typing be performed at all loci to which a recipient has detectable anti-HLA antibodies, using enough resolution (up to 2 field) to evaluate allele-specific antibody relevance.

DONOR DATA SUPPORTING CALCULATED PANEL REACTIVE ANTIBODY CALCULATORS

Foundational to the estimate of calculated panel reactive antibody (cPRA)[138,139] (see section later) is a database of many deceased donor HLA typings to which an antibody profile can be compared. It is essential that these data sets be large and as complete at all loci as possible so that the calculation

Table 69.3 Quantifying Human Leukocyte Antigens Versus Eplet Mismatches

Human Leukocyte Antigen Locus	A*	B*	Cw*	DRB1*	DQA1*	DQB1*	Total Mismatches
Recipient	01:01	07:02	07:01	03:01	01:03	02:01	
	03:01	08:01	07:02	13:09	05:01	06:03	
Donor	02:01	44:02	03:01	01:01	01:02	06:-2	
	11:01	54:01	05:01	04:05	03:01	03:02	
HLA antigen mismatch		6			4		10
Eplet mismatch		35			51		86
Recipient	01:01 24:02	51:01 35:01	07:01 01:02	03:02(18)	03:01	02:02	
				07:01	04:01	04:02	
Donor	23:01 36:01	52:01 53:01	15:02 18:01	03:01(17)	05:01	02:01	
				11:04	–	03:01	
HLA antigen mismatch		6			4		10
Eplet mismatch		4			23		27

HLA antigen mismatch is only a very crude estimate of the degree of mismatch between donors and recipients. In these examples, despite the same number of Class I and Class II antigen mismatches, the first pair has almost 3 times the number of eplet mismatches (not shown) representing a more potent alloimmune stimulus.

of cPRA is immunologically relevant and accurately representative of access.

HUMAN LEUKOCYTE ANTIGEN ANTIBODY SCREENING AND IDENTIFICATION

CONTEXT

Sensitization occurs when an individual is exposed to non-self HLA epitopes via pregnancy, prior transplant or blood transfusion. As a result of sensitization, a significant proportion of individuals can develop HLA antibodies: 50% to 74 % in pregnancy,[140–142] 1% to 20% of transfusion-exposed patients,[142–144] and 45% to 80% of patients with prior transplants.[145–148] A small percentage of patients also appear to have some HLA antibodies in the absence of a sensitizing event, but the clinical significance of these antibodies remains questionable.[149]

The detection of HLA antibodies prior to transplantation and their precise specificities (the exact antigens to which they are expected to bind) allows estimating the percentage of available donors to whom they may have DSA. Further, this allows estimating access to transplantation because of antibody profile. At the time of considering a potential donor, the recipient HLA antibody specificities can be examined for DSA to those of the donor in order to make a more informed and precise immunologic risk and transplant decision, as DSA portends a higher risk of humoral immune complications posttransplant.

After transplantation, the development of de novo DSA carries a particularly deleterious prognosis and is strongly associated with worse outcomes, both at the short and long term and with graft survival. Patently clear, however, is that both sensitive and specific antibody detection as well as identification methods are critical in guiding both the interventions and treatments for better outcomes, at all times during the transplant patient lifecycle.

ANTIBODY DETECTION AND IDENTIFICATION METHODS

SEROLOGY

The term *panel reactive antibody* (PRA) is broadly used to describe the percentage of donors in a candidate donor population to whom a recipient may have DSA. The term *PRA* comes from the original serologic methods for detecting HLA antibodies. PRA testing involves exposing a panel of various cells from different donors (selected to represent the common HLA frequencies in a potential donor population) to a recipient's serum, to which complement, and a vital dye are added. The percentage of the cells in the panel which are lysed is an estimate of the percentage of donors from that population against whom the recipient has cytotoxic antibodies (IgG1 or IgG3 of sufficiently high titer to initiate the complement cascade).[150–152] Due to the limitations of serologic screening, including lack of sensitivity, limited specificity with high false-positive rates due to non-HLA antibodies, IgM and autoantibodies[153–155] (which are immunologically irrelevant) and volatility of the PRA estimate based on cell panel phenotype variation regardless of recipient serum changes, this method will not be further discussed.

SOLID-PHASE HUMAN LEUKOCYTE ANTIGEN ANTIBODY SCREENING

In these assays, purified HLA antigens are covalently bound to inert beads or less commonly to an ELISA platform.[156–158] The latter have been largely replaced with bead-based assays due to their increased sensitivity and capacity for high-volume testing and will not be further discussed. In screening solid-phase assays, beads carry the complete class I or class II HLA profile of an individual and can be detected when the antibody in serum binds, regardless of complement activation, with a secondary fluorescent anti–human IgG detector antibody on a flow cytometric platform. A bead is determined to be positive when its fluorescence is above the threshold level validated by the laboratory and manufacturer. The PRA estimate, in this case stratified by class, is similar in principle to serology methods as the percentage of total beads with positive fluorescence.[159–167]

SOLID-PHASE HUMAN LEUKOCYTE ANTIGEN ANTIBODY-SPECIFICITY TESTING

Although PRA testing yields a population-based estimate of the number of donors to whom a recipient may have DSA, for these to be used in transplant decision making and donor selection, greater detail is needed as to which precise antigens the antibodies are directed toward. Single antigen beads (SABs) are similar in platform to HLA solid-phase screening assays, but in this case each bead has only one distinct type of HLA antigen bound. Also analyzed on flow cytometric platforms, up to 100 unique beads per class can be tested simultaneously.[168] The output of this test is also fluorescence-based and defines a list of antibody specificities that then must be reverse engineered into a cPRA using a calculator informed by large numbers of donor genotypes.[138,139,169]

COMPLEMENT-BINDING SOLID-PHASE ASSAYS

These assays are derived from the classic SAB test, and instead of simply detecting any antibody bound to the bead's antigens, they are designed to detect the binding of C1q or C3d complement by HLA-specific antibody.[170] Complement-binding DSA are associated with an acute rejection phenotype that is more severe and with an increased risk of allograft failure compared with noncomplement-binding DSA.[171] These assays will only be positive if IgG 1/3 antibodies are also first bound in sufficient density (high concentration) to antigens on the bead surface.[171–173] If low-titer IgG 1/3 antibody is present or IgG 2/4 isotypes dominate the serum antibody profile, then the result will be negative. Isotype-specific assays can also be used to discriminate between all four isotypes within a given SAB reaction and infer similar complement-binding potential, but are not commonly used in routine clinical practice.[174,175] It is notable that almost all antibodies that are detectable on traditional SAB platforms are present in all four isotypes,[176,177] suggesting that complement activation is at least possible for all antibodies, and its detection is dependent largely on titer of the IgG1/3 components.[170,173]

NON-HUMAN LEUKOCYTE ANTIGEN ANTIBODIES

Though not yet routinely implemented in all labs as a part of standard testing, commercial assays are now available for several non-HLA antibodies, including endothelial antibodies,[178,179] MICA,[61,68] and angiotensin type 1 receptor antibodies[88,97] in renal transplantation. These assays are also solid-phase–based with correspondingly similar methodologies as other solid-phase assays. Although there have been studies detailing univariate associations with transplant outcomes, the incremental additional value of these tests for population screening remains uncertain[179,180] and their current utilization potential is determined at the individual patient level.

INTERPRETING SOLID-PHASE HUMAN LEUKOCYTE ANTIGEN ANTIBODY TESTS

GENERAL INTERPRETATION

The numerical output from both screening and SABs is mean fluorescence intensity (MFI), which is compared with both negative control bead and negative control serum fluorescence in order to identify whether a given bead is of sufficiently high fluorescence to be deemed positive. It is important to note that although MFI output is numerical, it should not be interpreted as quantitative or an equivalent to antibody titer (see Limitations later).[181–183] Other factors can and should be considered in determining the relevance of a given bead reactivity, including sensitization history (and HLA type(s) of the sensitizer(s), if available), a pattern of shared eplets or a known Cross Reactive Group (CREG) among beads deemed to be fluorescing that would support their positivity as biologically plausible from an antibody-eplet binding perspective, a change in reactivity pattern over time (where a bead representing an antigen of interest and previously lower ranked in the assay MFI is now relatively increased, even if below usual thresholds for positive), interval changes in immunosuppression treatments, the presence of autoimmune disease with antibody binding to denatured antigens in the SAB assay[184] and the administration of any agents that could increase background interference. Once the determination of the list of positive beads has been made, further analysis of these positive reactions is required to determine the list of antibody specificities.

ANTIBODIES REPORTED AT THE ANTIGEN LEVEL

Antibodies are identified most commonly as specific to HLA antigens. These are reported as lists of serologic equivalents in nomenclature; for example, A2, B7, DR15, and so on. The interpretation by the clinician is correspondingly that the patient has an antibody that will bind all members of the antigen group regardless of specific allele. Biologically, it is inferred that the antibody is directed toward the epitope(s) that are common to all members of the antigen group and not at the epitope(s) that would distinguish unique alleles. Additionally, for class II, although not traditionally considered antigens in their own right, antibodies can form to the unique polymorphic alpha chains (coded by DQA1 and DPA1 genes). When these are identified, they will be listed with the alpha chain specified. For example, DQA1 or DPA2 antibody, distinct from DQ7 or DP3, which infers the antibody is to the beta chain.

ALLELE-SPECIFIC ANTIBODIES

Although antibodies are most commonly identified toward antigen groups, it is recognized that there are multiple different alleles comprising each antigen group. Sometimes, an antibody may be directed to one of the eplet/epitopes unique to a certain allele, in which case it will typically be identified using up to four digits; for example, B5102 antibody, where this nomenclature implies an antibody binding to the protein encoded by HLA-B*51:02 allele but not the other alleles in the B51 chain, including B*51:01 (Fig. 69.8).

ALPHA/BETA ANTIBODIES

Given the nature of class II molecules, it can occur that Fab' of the antibody can bridge the polymorphic alpha and beta chains of an antigen, resulting in an antibody that is specific to the unique combination of an alpha and beta chain. Wherever this occurs, the antibody is identified using some combination of the alpha and beta nomenclature; for example, DQA5-DQ2 (or similar).

CALCULATED PANEL REACTIVE ANTIBODY (OR EQUIVALENT) DETERMINATION

A recipient's list of antibodies is inherently useful in determining toward which donors a recipient has antibodies. However, to estimate a better likelihood of access it is useful to understand to what percentage of donors in a population the recipient bears antibody. To determine this, the list of antibodies is compared with a database of donor typings specific to the population of interest. The percentage of donors to whom a recipient has one or more DSA represents the cPRA[139,185] or equivalent (e.g., calculated reaction frequency or cRF in the United Kingdom).[186] The cPRA varies based not only on the list of antibodies but also, and more importantly, to the loci included in the donor typing and frequency of target HLA antigens in the donor population of interest. This, in turn, is strongly related to population-based racial differences.[187] At the time of this publication, cPRA calculators most widely available do not incorporate allele-specific antibodies.

LIMITATIONS OF HUMAN LEUKOCYTE ANTIGEN ANTIBODY TESTING

DEFINING A POSITIVE BEAD

Although the output of the SAB assay is a numerical MFI on each bead, which has a unique antigen on it, the MFI alone is insufficient to determine whether or not the fluorescence on a bead represents a true antibody being detected, and no universal threshold for positive should ever be assigned.[188] There is a widely varying density of HLA antigens between beads, such that the maximum MFI on any given bead may be determined by antigen density rather than serum level of antibody. In the presence of very high levels of complement-binding antibody; C1q may be generated in vivo from soluble complement proteins and interfere with the binding of the antibody to the bead, significantly lowering the MFI of the beads, representing very high levels of antibody. Dilution of serum samples when this phenomenon occurs can clearly show that MFI increases with serum dilution, confirming its nonquantitative value on the undiluted serum. Very commonly, groups of HLA antigens share common epitopes to which antibodies are targeted, and when the antibody to a shared target is present at a subsaturating level, it can spread across the multiple beads with antigens sharing this target, thereby reducing the MFI of any given single

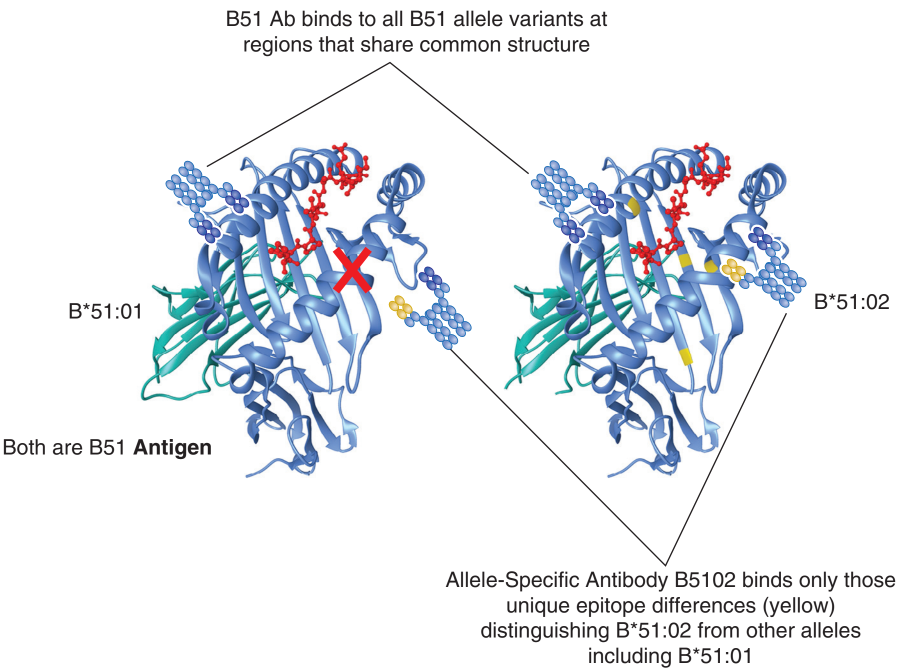

Fig. 69.8 Allele differences in protein structure can lead to different immunogenicity. Proteins coded by B*51:01 and B*51:02 alleles differ by a 3 amino acid substitution in the peptide-binding region. Both are referred to as B51 antigens. Antibodies that bind to regions common to all the B51 antigens will bind proteins coded by all alleles in the group. Some antibodies bind only to those specific epitope region(s) that distinguish one allele from the others. These allele-specific antibodies may be identified by their four-digit names, in this example, B5102 antibody.

Clinical Relevance

Although the output of the single-bead assays is a numerical mean fluorescence intensity on each bead, which has a unique antigen on it, the mean fluorescence intensity alone is insufficient to determine whether or not the fluorescence on a bead represents a true antibody being detected and no universal threshold for positive should be assigned. For multiple lab-specific validation and reagent-based reasons, it is strongly recommended that the readers consult with their own laboratory as to the analytic strategies employed to optimally identify antibodies and thresholds used for calling a positive test.

bead within the group. This single-bead MFI can be misleadingly low as an estimate of antibody. Similarly, some antigens that have several common alleles are represented on more than one bead in the SAB assay; again, an antibody to this antigen at an epitope shared by all alleles can be diluted across the multiple beads, lowering single-bead MFI estimate. For these and other lab-specific validation and reagent-based reasons, it is strongly recommended that the readers consult with their own laboratory as to the analytic strategies employed to optimally identify antibodies and thresholds for positive (Fig. 69.9).[181–183]

DEFINING IMMUNOLOGIC RELEVANCE

A conundrum of the detection of antibody, as for all laboratory results, is that mere identification does not precisely determine relevance at that time or in the future. Whereas higher-titer antibodies, persistent (versus transient) antibodies, and class II antibodies may be associated with worse outcomes post-transplant, it would be misleading to classify these as absolute risk states. Rather the clinical relevance of an antibody is part of a multifactorial patient and donor assessment in the context of patient history, immunosuppression, sensitizing events, and current clinical status and organ function. There are no indisputable features that clearly discriminate antibodies with more deleterious or immediate consequences from those that portend a more benign course.

NATURALLY OCCURRING ANTIBODIES

SAB assays are so sensitive that reactions may be detected in individuals without any history of sensitizing events.[149,184] The clinical consequences of these antibodies, the nature of their target antigen (denatured in assay versus intact in vivo) and whether they should be avoided or considered in donor selection still requires further study.

ANTIBODY CAN BE BOUND TO THE ALLOGRAFT

The SAB assay detects circulating antibodies. However, it has been shown, in individuals with a failed allograft in place, that significant antibody may be removed from circulation and bound to the endothelium, reducing the sensitivity of the assay. Following nephrectomy, antibody levels may increase.[52,189] If an allograft remains in situ, the full antibody potential may be dampened.

COMPLEMENT FIXING ASSAYS

Although some studies have shown that antibodies detected by complement fixing assays are associated with worse

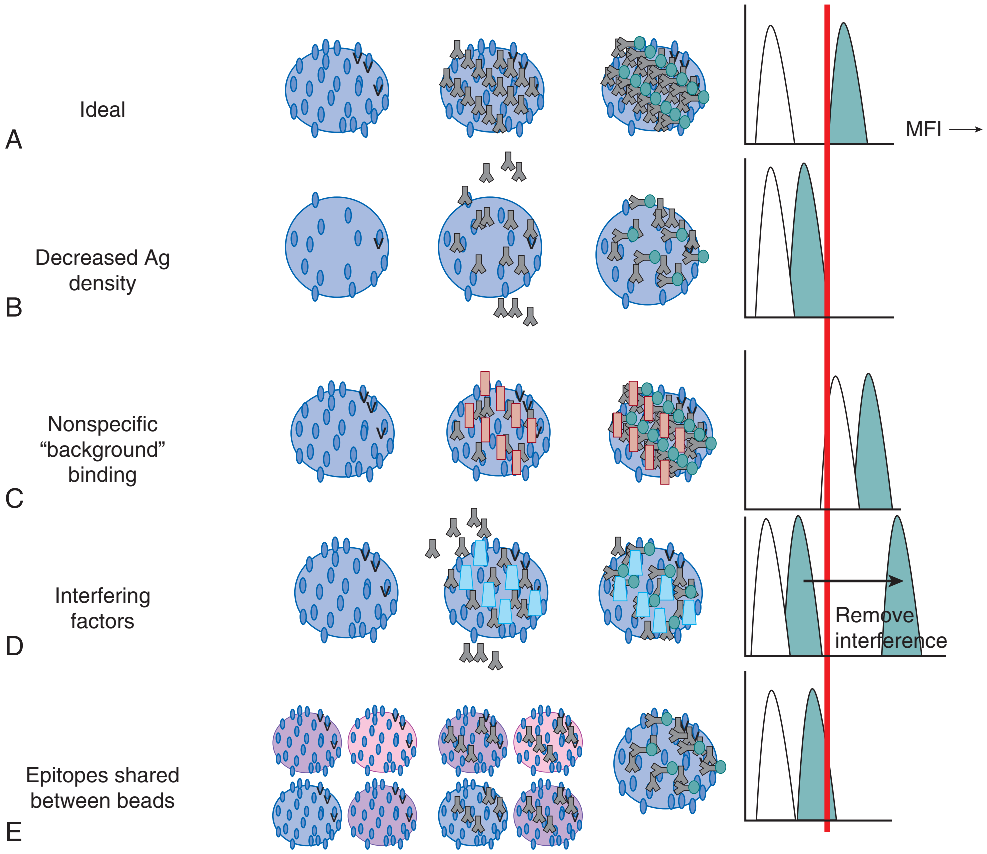

Fig. 69.9 Median fluorescence intensity (MFI) of single antigen bead assays has analytic limitations and cannot be used as a quantitative metric of antibody amount. A, An ideal test should always be able to distinguish antibody binding *(blue signal)* from negative control *(white)* with a clear threshold and no overlap between the MFI distributions. B, Decreased density of antigen (Ag) on the surface of the bead will result in MFI measurement that underestimates the amount of the antibody present. C, In contrast, nonspecific binding to the bead can result in artificially high background and signal MFI, with overestimation of antibody. D, Interfering substances may prevent the detection of the antibody of interest with lower MFI. E, Epitopes shared between different beads can dilute the amount of antibody bound to any single bead, with an erroneously low MFI on the given bead of interest. (From Konvalinka A, Tinckam K. Utility of HLA antibody testing in kidney transplantation. *J Am Soc Nephrol* 2015;26(7):1489–1502. With permission.)

outcomes, not all studies demonstrate this, and pretransplant routine SAB testing alone predicts posttransplant outcomes as well as complement assays. These assays do not detect specific deleterious antibodies, but rather higher-titer IgG1/3 in vitro.[172,190] If there are multiple low-titer IgG 1/3 antibodies in a sample, they may be C1q negative in vitro, but when present in vivo, they may nonetheless activate complement on the endothelium. Indeed, C1q positivity does not correlate with C4d presence in tissues,[191] and merely concentrating a C1q-negative serum can change results to C1q positive.[173] Finally, not all deleterious antibody-mediated pathways are complement-dependent; it is erroneous to surmise that antibodies that do not bind complement in vitro have less potential for harm in vivo.[170]

GENERALIZABILITY OF CALCULATED PANEL REACTIVE ANTIBODY

For a cPRA to best estimate the percentage of potential donors to whom a recipient has DSA (and consequently higher immune risk to those donors), not only do the antibodies need to be accurately identified, but also the donor typings utilized must be representative of the donor population accessible to that individual. As such, a cPRA based on one donor population may not be an accurate assessment of donor access in other populations with different racial distribution and different associated HLA antigen frequencies.[187] Additionally, the cPRA may be systematically underestimated when donor typing is not complete at loci where HLA antibodies to those loci are utilized in donor decision making.[139]

USING HUMAN LEUKOCYTE ANTIGEN ANTIBODY TESTING IN KIDNEY TRANSPLANTATION

ON THE WAITLIST

In addition to HLA typing of the recipient, waitlist HLA antibody testing is foundational to ongoing patient immunologic assessment. The main goal of waitlist testing is to estimate the transplant candidate's access to donors from the perspective of avoiding donor antigens to whom they have antibodies. This estimate is quantified as the cPRA, where (100-cPRA) is the percentage of donors otherwise medically and ABO suitable that would be accessible to the recipient. If the candidate has a high cPRA, longer wait times and higher waitlist mortality are anticipated.[192–194] In that case, strategies to improve access to more donors should be sought. If no potential living donor is available, then desensitization,[195–197] or deceased donor acceptable mismatch allocation strategies,[198–200] should be considered. If a living donor is available, then desensitization or kidney paired donation programs[201] could improve access. A second goal of waitlist antibody testing is to compile a longitudinal immunologic profile of the candidate; antibodies can wax and wane over time, so repeated testing is indicated (usually every one to three months) to ensure the most complete list of antibody potential is compiled for use at the time of a potential donor evaluation. If a patient incurs a sensitizing event, repeat testing 6 weeks after the event is recommended to capture any memory responses or de novo antibodies. It is critical to remember that patients frequently have a long immunologic history predating their entrance to the waitlist, so even repeated HLA antibody testing may not identify all antibody potential; a thorough history of sensitizing events both prior to and while on the waitlist remains foundational to counselling patients as to their immunologic risk posttransplant. Finally, the cPRA and its contributing antibody specificities are, in and of themselves, not a measurement of immunologic risk (only risk of reduced access to donors); immunologic risk is a measurement unique to each donor-recipient pair under consideration.

At the Time of Transplant Offer: Virtual Crossmatching

In conjunction with donor HLA typing, the full history of recipient HLA antibodies may be compared with donor HLA antigens in what has come to be known as a virtual crossmatch (VXM). If any of the current or historical specificities correspond to donor antigens, these DSA confer a positive VXM. Under optimal circumstances, the VXM is intended to be a predictor for an actual cell-based crossmatch performed on the corresponding sera, in order to facilitate allocation and transplant decision making. However, the predictive values of VMX for actual crossmatch (XM) can vary widely depending on XM methodology[202–204] and the laboratory- and program-specific parameters for identifying antibodies. The current VXM reflects the most recent serum tested prior to donor offer and most closely reflects the antibodies likely to be present at the time of transplant and is associated with early and late rejections and reduced allograft survival. Conversely, the historical VXM reflects the cumulative history of DSA. When current VXM is negative but historical is positive, there is an increased risk for memory B-cell reactivity after transplant and corresponding adverse allograft outcomes. Despite historical and fewer recent reports of high PRA association with adverse transplant outcome, the ability to more precisely assess VXM/DSA has revealed that it is in fact DSA that drives adverse outcomes, regardless of the level of PRA directed toward third-party antibody, further improving treatment decision making.[200,205,206] In some jurisdictions where regulatory requirements allow, the VXM has replaced the actual XM for many transplant decisions, with comparable transplant outcomes.[207,208] Additionally, the virtual crossmatch is an important part of crossmatch interpretation to ensure that the cell-based crossmatch, if positive, is in fact immunologically relevant[165] (see Crossmatching later).

POSTTRANSPLANT ANTIBODY MONITORING

Following transplantation, all HLA mismatched patients are at risk of de novo HLA DSA development. Landmark studies in 2005 and 2007 first identified that posttransplant HLA antibodies had higher rates of graft failure than those without antibody development.[68,69] In the years since, it has become clear that *de novo* HLA DSA development is a major risk factor for rejection and premature graft loss. Despite numerous studies, there is no consensus to date as to the optimal timing and frequency of monitoring.

Although associations of de novo DSA and the range of transplant-associated outcomes cannot be absolute or perfectly predictive, de novo DSA remain a strong independent risk factor for significant antibody-mediated outcomes. De novo DSA appear at a median of 3 to 68 months posttransplant and positive de novo DSA are found in 6% to 38% of patients 3 years after transplantation.[122] Class II antibodies predominate in most series. As a diagnostic tool the presence of DSA is supportive of a diagnosis of ABMR in the context of aligned pathology and dysfunction,[53,209–211] but ABMR may also occur in the absence of detectable HLA DSA, often dependent on timing of testing, with antibody bound in the failed allograft[52,189] or rarely to non-HLA antibody.[51,61,212] Late ABMR is associated with chronic pathology, Class II Ab and lower response rate to treatment.[213–215] DSA MFI thresholds do not reliably discriminate prognosis or treatment responsiveness,[213,216] nor is it a tool to determine duration of treatment as DSA frequently persist well beyond clinical and histopathologic resolution of ABMR.[214,216]

As a prognostic tool, when detected before graft dysfunction or pathology of ABMR, the role of de novo DSA role is not clearly defined. The interval between de novo DSA detection and clinical outcome can vary widely from months to years, as can the phenotype of the outcome itself, suggesting diverse pathways of injury at play with many effect modifiers.[191,217–223] Data do not presently support empirical treatment for de novo DSA in the absence of graft dysfunction and/or tissue injury. At this juncture, asymptomatic de novo DSA can best be considered a higher-risk state for future adverse events that may warrant closer patient follow-up.

DONOR-SPECIFIC (CELL-BASED) CROSSMATCHING

CONTEXT

Before the introduction of the complement-dependent cytotoxicity (CDC) crossmatch in 1969,[100] the phenotype of early graft loss (as hyperacute or accelerated rejection) was a recognized but largely unpredictable event. The introduction of the CDC crossmatch provided clinicians with a tool to predict early graft loss prior to transplantation and improved decision making. The cell-based crossmatch in general may be considered a "surrogate" transplant; rather than exposing the endothelial cells of the allograft to the recipient circulation at the time of operation, the donor lymphocytes (expressing the same HLA antigens as donor endothelium) are first exposed to recipient serum in vitro. Where HLA antibodies are detected on the surface of the lymphocytes, it is surmised that these antibodies would also bind to the allograft endothelium with deleterious consequence; the transplant can then be avoided, or immunosuppression modified. Although supplanted by

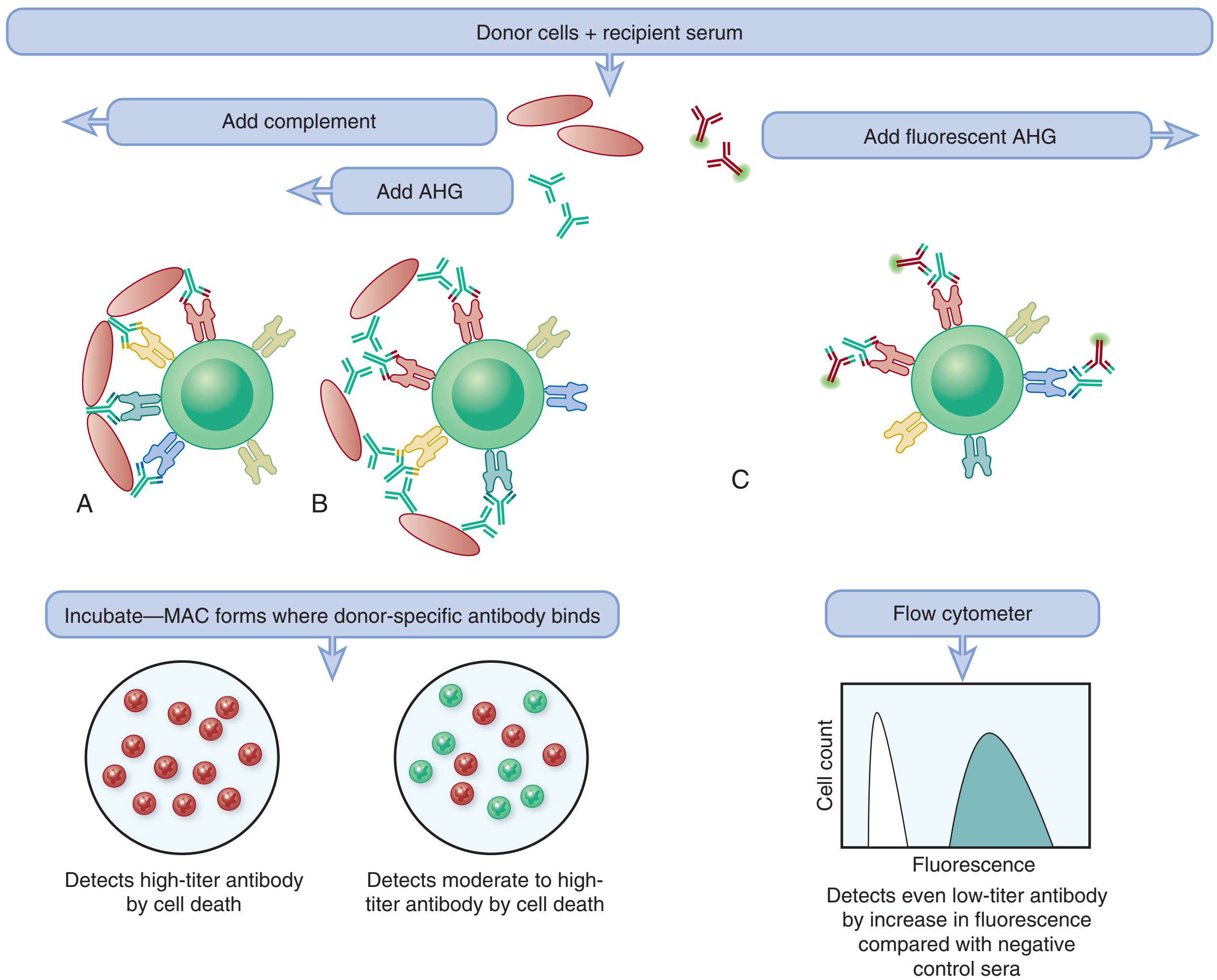

Fig. 69.10 Crossmatch methods. A, The complement-dependent cytotoxicity (CDC) crossmatch detects high-titer donor-specific antibody (DSA) when it binds to the cell in sufficient density to activate the complement cascade. The resulting formation of the membrane attack complex (MAC) results in cell death, which can be detected via microscopy with a vital dye. A positive CDC crossmatch due to human leukocyte antigen (HLA) antibody is associated with hyperacute or early accelerated rejection. B, The antihuman globulin (AHG)-enhanced CDC crossmatch increases sensitivity for moderate-titer antibody with the addition of AHG. AHG binds to donor-specific antibody (DSA) already bound to its target antigen, allowing for activation of complement at lower levels of original DSA. The MAC will form again and kill cells with antibody bound to it, deleted under microscopy. A positive AHG-CDC crossmatch due to HLA antibody is associated with hyperacute or early accelerated rejection. C, The flow cytometry crossmatch can detect even lower-level antibodies using fluorescence detection if the antibody is bound to the cell. Noncomplement-binding antibodies are also detected using this methodology. Positive flow cytometry crossmatches may vary in strength. Strongly positive results due to HLA antibodies are associated with not only higher rates of earlier antibody-mediated rejection (AMR) but also subclinical rejection and chronic AMR. However, not all positive flow cytometry crossmatches are clearly associated with an adverse outcome, and methods to distinguish pathologic from nonpathologic antibodies are needed. (From Lapointe I, Tinckam K. *Histocompatibility testing for kidney transplantation risk assessment*. Scientific American Nephrology, Dialysis and Transplantation. ©Decker Intellectual Properties, Inc. 2017. With permission.)

virtual crossmatching in some clinical circumstances, the cell-based crossmatch remains an important test for donor recipient compatibility immediately prior to transplant.

CROSSMATCH METHODS

COMPLEMENT-DEPENDENT CYTOTOXICITY

This method employs the same serologic principles described for the earliest typing and antibody detection methods, where cells from the donor are mixed with recipient serum in the presence of complement and cell death detected by vital dye uptake is the marker of a positive test, detecting high-titer complement-binding antibodies likely to result in hyperacute rejection (Fig. 69.10).

ANTIHUMAN GLOBULIN ENHANCED COMPLEMENT-DEPENDENT CYTOTOXICITY

By the early 1990s a more sensitive derivation of the CDC crossmatch was achieved by adding antihuman globulin (AHG) to the test, binding any antibody on the cell surface, increasing overall antibody density and increasing the probability that complement would become activated with lower levels of HLA antibody. These lower-titer HLA antibodies were nonetheless clinically relevant and associated with earlier antibody-mediated outcomes not predicted by the less sensitive CDC crossmatch.[224–229]

FLOW CYTOMETRY CROSSMATCH

With the awareness that antibody-mediated adverse outcomes could still occur with negative AHG-CDC crossmatch, the flow cytometry crossmatch (FCXM) permitted detection of even very low-level and/or noncomplement-binding antibody on the cell surface, detected by fluorescent anti–human IgG rather than complement activation.[230–235]

INTERPRETING CELL-BASED CROSSMATCH RESULTS

Crossmatches are performed separately on both T and B lymphocytes. T lymphocytes express Class I HLA antigens whereas B lymphocytes express both class I and Class II HLA antigens. Historically, it was thought that only the T-cell crossmatch representing Class I HLA antigens predicted antibody-mediated early outcomes, but it is now known that endothelium that has sustained injury (for example, ischemia-reperfusion as is common in transplantation) can also upregulate class II expression so that positive B-cell crossmatches, when due to HLA antibody, can result in early accelerated rejection even when the T-cell crossmatch is negative.

A positive cytotoxic crossmatch when due to HLA antibody, (see Limitations later), indicates high-titer complement-binding antibody and predicts hyperacute rejection. A positive AHG-CDC XM (with negative CDC XM) detects lower-titer, still complement-binding antibody, and may be associated less commonly with hyperacute rejection, and more commonly with early or accelerated rejection in the hours to days after transplant. Positive FCXM, on the other hand, detects both complement and noncomplement-binding antibodies, and where cytotoxic crossmatches are negative and tend to be associated with later rejection and graft loss outcomes, they are less likely to result in hyperacute or accelerated rejection as well.

LIMITATIONS OF CELL-BASED CROSSMATCH

COMPLEMENT DEPENDENCE FOR DETECTING HUMAN LEUKOCYTE ANTIGEN ANTIBODIES

The use of the CDC XM greatly reduced, but did not eliminate, early accelerated rejection (false-negative test),[236] which could result from antibodies below the level of detection by complement-dependence that still caused damage to the graft endothelium in the hours to days after transplantation.

IMMUNOLOGICALLY IRRELEVANT NON–HUMAN LEUKOCYTE ANTIGEN TARGETS ON CELLS

False-positive results can occur from autoantibodies[150,153–155] or immunologically irrelevant antibodies (including IgM antibody) binding non-HLA targets.[155,207,237,238]

DRUG INTERFERENCE

Drugs that bind to lymphocytes, such as Thymoglobulin, Alemtuzumab and Rituximab or that contain nonspecific antibodies (intravenous immunoglobulins) may result in false-positive XM.

B-CELL FALSE-POSITIVE CROSSMATCH

The B-cell receptor includes constitutively expressed immunoglobulin that can increase the likelihood of a positive B cell crossmatch in the absence of donor-specific HLA antibody. Laboratories routinely account for this by setting cutoffs for positivity and by using Pronase in flow cytometry crossmatching, which can improve the crossmatch specificity.[239–243]

INCREASING SENSITIVITY MAY REDUCE PREDICTIVE VALUE

With all methods that increase sensitivity, the strength of correlation with any clinical outcome of interest becomes less specific. Although higher rates of adverse outcomes occur with positive FCXM in isolation, a positive FCXM does not guarantee worse transplant results, and in some cases these transplants can be performed with short- to medium-term success in the context of augmented immunotherapy or desensitization.

LABORATORY VARIABILITY

The cell-based crossmatch is not a commercially available assay but rather developed within HLA laboratories frequently using nonstandard incubation times and variable reagents and controls. As such, there are no standard cutoffs for positive results and considerable inter laboratory variability may occur at the lower limit of detection of antibody.[244]

USING THE CELL-BASED CROSSMATCH IN THE MODERN ERA

The cell-based crossmatch (most commonly the FCXM in the modern era) remains in many programs the gold standard for immediate pretransplant decision making. Unlike single antigen bead assays, which may not be easily performed on serum collected on the day of transplantation, the flow crossmatch may be quickly performed with serum that is temporally reflective of the immediate transplant environment. Considering the limitations described previously, we encourage the simultaneous performance of autocrossmatches (to assess for autoantibody) as well as crossmatches on historical sera that have been thoroughly evaluated using solid-phase antibody platforms; the authors strongly recommend that optimal interpretation of a crossmatch includes the solid-phase antibody data to confirm or exclude HLA antibody as the cause of the positive crossmatch. This is more easily arranged for living donor transplants; for deceased donor transplantation, historical sera crossmatches with their corresponding solid-phase antibody data that are already known may provide additional context for interpretation of a positive current serum cell-based crossmatch, where a current solid-phase result is not immediately available.

If a positive cell-based crossmatch is interpreted as caused by an autoantibody or non-HLA antibody, the transplant may proceed in most circumstances, as the crossmatch is deemed immunologically irrelevant.

If a positive cell-based crossmatch is interpreted as caused by an HLA antibody or an HLA antibody cannot be reliably excluded, transplant may be relatively contraindicated or require significantly augmented immunotherapy.

It is now true that as the VXM has become more commonplace, it has replaced the pretransplant cell-based crossmatch in some centers, where regulations permit, without demonstrable negative impact on patient and graft outcomes, and where VXM results are unambiguous in the context of a reliable history excluding recent sensitizing events. Nonetheless, in circumstances where the antibody

Table 69.4 Clinical Histocompatibility Testing Menu and Associated Interpretations and Actions

Timepoint	Clinical Question(s) Applicable	Tests Used	Interpretation and Potential Actions
Pretransplant	How sensitized is the patient? How difficult is it be to find a (HLA-based) suitable donor?	• HLA typing • HLA AB Screening and Identification	If reduced access to donors from high cPRA, then access to more donors is optimal (increased sharing for acceptable mismatches, kidney paired donation programs) or desensitization to reduce current burden of circulating antibody.
	What is the patient's risk of a memory response post-transplant?	• HLA typing • HLA AB screening • In the context of a good clinical history of sensitizing events	All patients with HLA antibodies as well as those with sensitizing events should be considered at higher risk of memory responses post-transplant; closer follow-up posttransplant may be warranted.
At Transplant	Is the patient PREDICTED to have HLA antibodies to the donor? Will the cell-based crossmatch prior to transplant be positive?	• The virtual crossmatch: Donor typing PLUS • HLA AB identification (current and historical results)	Positive HISTORICAL VXM: Patient is at higher risk of memory response. Some programs do not transplant; others augment immunotherapy or monitoring. Positive CURRENT VXM: Patient is at higher risk of ABMR. Transplant decision making depends on strength of antibody (AB test plus cell-based crossmatch).
Posttransplant	Has the patient developed de novo HLA antibodies? (no graft dysfunction)	Donor typing from transplant Posttransplant AB testing	If positive, may indicate risk of future or current subclinical) ABMR? Guides further diagnostics or monitoring noting certain HLA loci portend a worse prognosis (Class II).
	Does the patient have ABMR? (with or without graft dysfunction)	Donor typing Posttransplant AB testing In context of biopsy if available	HLA antibodies are supportive of a diagnosis of AMR but may not always be present.

AB, antibody; *ABMR*, antibody-mediated rejection; *HLA*, human leukocyte antigen; *VXM*, virtual crossmatch.

testing is indeterminate, the cell-based crossmatch still may serve as additional information to adjudicate the relevance of the antibody result.

An additional conundrum exists when the cell-based crossmatch is negative but the VXM is positive, due to the increased sensitivity of SAB over cellular assays.[245–247] ABMR has been reported in up to 55% of these patients[207,246,248,249] (greater than VXM negative counterparts) impacting graft survival in some,[249,250] but not all[195,251–253] series.

Indeed, the decision to proceed with transplant in the face of a positive crossmatch or negative crossmatch in the presence of HLA DSA should usually consider center expertise and the patient-specific competing risks of not receiving the transplant (including chance for a future DSA negative offer, and risk of dialysis-associated mortality).[196,254] The exception is with the positive cytotoxic crossmatch, where HLA DSA are confirmed because this confers an unacceptably high risk of hyperacute rejection precluding immediate transplant.

THE EVOLVING ROLE OF HISTOCOMPATIBILITY CONSULTATION

Over the almost 50-year history of histocompatibility testing, testing has gone from simple pretransplant serologic-based assays with dichotomous risk classifications guiding the most basic of transplant decision making to sensitive, specific, individualized, and dynamic testing throughout the duration of the transplant experience (Table 69.4). Indeed, the test platforms and their corresponding results have evolved to such a degree of complexity that significant and discipline-specific postgraduate training and expertise is required to render a thoughtful and informative opinion. Adding to the complexity is the awareness that all biologically based HLA testing has inherent strengths and limitations, and the understanding of the interpretation of these myriads of tests in the context of these limitations is highly nuanced and requires consideration of not only immunologic but also competing medical, surgical, infectious, and even socioeconomic risks to the patient.[250,255] Histocompatibility testing has evolved to such a state of complexity and expertise that ongoing patient-specific consultation with laboratory specialists beyond simply the passive receipt of testing results is a mandatory part of good clinical transplant care, pre, post and long after transplant has occurred.

KIDNEY ALLOGRAFT REJECTION

ACUTE CELLULAR AND ANTIBODY-MEDIATED REJECTION

With the advent of modern immunosuppressive regimens[256] and increased sensitivity of pretransplant anti-HLA screening,[257] the incidence of acute rejection has decreased substantially over the last few decades. Approximately 15%

to 20% of kidney transplant recipients now experience a first rejection episode in the first 5 years after transplantation.[258] Rejection episodes are classified histologically using the Banff classification system.[54] This system provides criteria for the diagnosis of acute cell-mediated and antibody-mediated rejection, which can coexist. The main histologic feature of acute T-cell–mediated rejection (TCMR) is a mononuclear leukocyte infiltrate located in the tubules, interstitium, and/or arteries of the graft.[54] The cellular infiltrate is classically composed of T cells and monocytes/macrophages. Acute cell-mediated rejections are classified as Banff grade 1A or 1B, depending on the extent of inflammation, when the cellular infiltrate affects only the tubules and the interstitium. When endarteritis is present, rejections are classified as Banff grade 2A, 2B, or 3 according to the severity of the vascular involvement. In acute cell-mediated rejection, vascular involvement has classically been regarded as portending a poor prognosis in terms of resistance to treatment and graft survival.[259] This may in part be due to the previously unrecognized coexistence of cell-mediated endarteritis with classical features of ABMR, such as DSA and microcirculatory inflammation.[260] In line with this observation, a recent cohort study showed that endarteritis did not increase the risk of graft loss compared with tubulointerstitial involvement alone in patients who were DSA negative.[261] In contrast, patients who had endarteritis and DSA experienced a risk of graft loss that was 9 times greater than that observed in patients with DSA-negative tubulointerstitial rejection.

In addition to the presence of circulating DSA or antibodies against other donor antigens, ABMR is characterized by acute tissue injury, as evidenced by either microvascular inflammation (glomerulitis, peritubular capillaritis), endarteritis, or, in the absence of other causes, thrombotic microangiopathy or acute tubular injury. Evidence of antibody interaction with the endothelium has long been recognized by diffuse C4d deposition in peritubular capillaries. However, as antibodies can target the endothelium through both complement-dependent and -independent cytotoxicity, the existence of C4d-negative ABMR has been well established in recent years.[53,262,263] Other markers of endothelial interaction with antibodies, such as increased endothelial gene transcripts[264] in the biopsy and at least moderate microvascular inflammation, are now also recognized as histologic features of ABMR.[54]

The use of microarray technology to analyze RNA transcripts from biopsies obtained from transplant rejection cases has provided novel insights on the mechanism of acute rejection in recent years. For instance, transcripts from NK cells and macrophages, as well as endothelial-associated transcripts, have been associated with DSA and ABMR.[48,264] Archetypal signatures of TCMR and various stages of ABMR have been found to correlate with graft survival and clinical judgment through transcript analysis, making this technology a promising tool to overcome some of the limitations of histologic analysis, such as poor reproducibility.[265,266] The clinical aspects of acute rejection are covered in detail in Chapter 70.

CHRONIC ALLOGRAFT INJURY

The term chronic allograft nephropathy or, more recently, *chronic allograft injury* (CAI) has been used to describe the clinical presentation of relatively slow renal function decline, proteinuria, and/or hypertension occurring more than 3 months after renal transplantation, as accompanied by interstitial fibrosis and tubular atrophy (IFTA) on the graft biopsy.[267] Because these features are not specific, kidney graft biopsy should be performed to investigate the cause of deteriorating graft function.[268] Recent data show that specific causes for IFTA can be found in over 80% of cases through biopsy and/or clinical history.[269,270] Causes include recurrent glomerular disease, transplant glomerulopathy, polyomavirus nephropathy, and pyelonephritis.[269,270] Although chronic active TCMR or ABMR can result in IFTA, their diagnoses require other histologic criteria that are presented later.

Rather than being a disease per se, CAI and IFTA are the end result of both alloimmune and nonalloimmune injuries that can occur in transplanted kidneys. Moderate to severe IFTA is a predictor of adverse graft outcomes, being associated with both impaired function and poorer graft survival.[271,272] Studies performed in the early 2000s pointed to an important role for calcineurin inhibitor toxicity in the development of IFTA. A study of cyclosporine-treated pancreas-kidney transplant recipients showed moderate to severe IFTA and arteriolar hyalinosis in 66% and 90% of surveillance biopsies, respectively, at 5 years.[273] However, recent data show that the incidence of IFTA is lower among solitary kidney transplant recipients, with only 17% of 5-year surveillance biopsies showing moderate to severe IFTA, and no significant differences observed between patients maintained on calcineurin inhibitors versus those who were not.[274] Hence although calcineurin inhibitors are profibrotic and nephrotoxic, their contribution to the progression of IFTA, relative to other causes of graft injury, is unclear.[275]

One of the key factors that trigger the development of IFTA is inflammation. Processes such as acute and chronic ABMR and TCMR, recurrent glomerular diseases, BK polyomavirus, CMV nephritis, and bacterial acute or chronic pyelonephritis all involve the activation of macrophages and graft infiltration by leukocytes (mainly T cells). Under the influence of proinflammatory cytokines, T cells, macrophages, and tubular epithelial cells produce TGF-β, connective tissue growth factor (CTGF), and other profibrotic cytokines. In response to these cytokines, fibroblasts, fibrocytes and pericytes are activated and transform into matrix-producing contractile myofibroblasts, leading to allograft fibrosis.

Another important trigger of fibrosis is microvascular injury, which can occur during ischemia-reperfusion or acute and chronic ABMR. In kidney transplant patients, ischemia-reperfusion injury occurs at the time of transplantation. Acute kidney injury that occurs in the immediate posttransplant period is termed *delayed graft function*. Delayed graft function is associated with a moderately increased risk of graft loss[276] and with the degree of IFTA after transplantation.[275] In experimental models of ischemia-reperfusion injury, renal perfusion in peritubular capillaries is compromised within minutes of unclamping.[277,278] Endothelial dysfunction/injury and apoptosis compromise microcirculatory renal blood flow through decreased vasodilatory capacity, coagulation activation and the formation of microvascular thrombi, and increased rolling/adhesion of inflammatory cells.[279,280] Because the regenerative capacity of endothelial cells in peritubular capillaries appears limited, microvascular damage occurring during an episode of acute kidney injury can lead to permanent peritubular capillary rarefaction. Loss of peritubular capillaries favors chronic hypoxia, leading to overexpression of hypoxia inducible factor 1-α (HIF-1α). HIF-1α promotes the transcription of fibrogenic genes, such as TGF-β and CTGF, accumulation of α-smooth muscle actin-positive myofibroblasts, and the production of fibrogenic mediators by dying endothelial cells.[281] In transplanted kidneys, the magnitude of microvascular involution during the first 3 months following transplantation is a major negative predictor of long-term renal allograft function.[282] Recent studies using in vivo imaging and electron microscopy demonstrated a tight correlation between peritubular capillary dysfunction/rarefaction and renal fibrosis in both murine models of acute kidney injury and human kidney biopsy samples.[282–287]

CHRONIC ACTIVE CELLULAR AND ANTIBODY-MEDIATED REJECTION

Allograft vascular involvement by cell- or antibody-mediated processes is the hallmark of chronic active TCMR and ABMR. Chronic allograft arteriopathy, defined as arterial intimal fibrosis with mononuclear cell infiltration and neointima formation, is a key feature of chronic active TCMR, but is also seen in chronic active ABMR.[54] Other histologic criteria for chronic active ABMR include the presence of circulating DSA, evidence of chronic vascular damage and remodeling of the graft microcirculation (double contours in glomerular capillaries or multilayered basement membrane in peritubular capillaries), diffuse C4d staining in peritubular capillaries, and gene transcripts of endothelial injury.[54] The presence of inflammation in zones of IFTA is associated with decreased graft survival,[288,289] and may be seen in chronic active TCMR.[54] The clinical aspects of chronic active ABMR and TCMR are discussed in Chapter 70.

MECHANISMS OF ACTION OF IMMUNOSUPPRESSIVE DRUGS

To prevent the development of rejection, immunosuppressive drug regimens comprise induction and maintenance immunosuppressive agents. Induction agents are powerful immunosuppressants that are administered in the immediate perioperative period and discontinued in the early posttransplant period. These agents include polyclonal antibodies (anti-thymocyte globulins or ATG) and monoclonal antibodies (basiliximab, alemtuzumab). Maintenance immunosuppressive agents are also administered at the time of transplantation but are continued for the lifespan of the graft and sometimes beyond. They include calcineurin inhibitors (tacrolimus and cyclosporine), mycophenolate mofetil, rapamycin, azathioprine, and corticosteroids. The majority of American centers are currently using ATG (60%) or basiliximab (20%) for induction, and a combination of tacrolimus, mycophenolate mofetil, and corticosteroids for maintenance.[290] The first-line treatment of acute TCMR is steroid pulses, while ATG and alemtuzumab can be used as second-line therapy for steroid-resistant rejection. Some centers use antibodies in combination with corticosteroids as first-line therapy for Banff grade 2 TCMR or higher.[291] The clinical aspects of acute rejection management are covered in more detail in Chapter 70. Here, we will discuss the mechanism of action of immunosuppressive agents. Fig. 69.11 illustrates the action mechanisms of the most frequently used immunosuppressive drugs.

INDUCTION AGENTS

POLYCLONAL ANTIBODIES

Polyclonal ATG are produced by injecting animals, such as horses or rabbits, with human thymocytes to obtain the purified γ-globulin fractions of the resulting immune sera. Many preparations have been developed, marketed by different companies, but the one that is most commonly used in clinical practice is rabbit antithymocyte globulin (rATG-Thymoglobulin, Genzyme). Two other existing preparations are rATG-Fresenius, where rabbits are immunized with Jurkat T-cell leukemia lines instead of human thymocytes, and equine ATG (ATGAM, Pfizer), which is a result from the immunization of horses with human thymocytes. Because polyclonal antibodies originate from animals, antirabbit (or antiequine) antibodies can form after ATG administration, which can lead to decreased activity and serum sickness upon re-exposure.[292]

ATG targets multiple epitopes present on thymocytes, including immune response antigens and adhesion/cell trafficking molecules, which explains its various mechanisms of action.[293] Although its primary mode of action is T-cell depletion, ATG can also bind to B cells, DCs, and nonlymphoid cell lines, such as monocytes, neutrophils, platelets, and red blood cells.[294] ATG induces lymphocyte depletion in the peripheral blood primarily by complement-dependent cell lysis, although other mechanisms, such as ADCC and induction of apoptosis through Fas/Fas ligand interactions, have been documented.[294–296] Lymphocyte depletion also occurs in the spleen and axillary lymph nodes, mainly through T-cell apoptosis.[294] Although the T-cell count returns to baseline 10 days after a single dose of ATG,[297] about 40% of patients treated with a full course recover more than 50% of the initial lymphocyte count after 3 months.[298]

Another important mechanism of action of ATG is modulation, which occurs both in the peripheral blood and peripheral lymphoid tissues.[294] ATG binds to antigenic target molecules on the cell surface, including the TCR/CD3 complex. The antigen–antibody complex is then internalized, and the pathway through which the target cell surface molecule was acting remains inhibited for up to 4 weeks. This downmodulation affects not only molecules that regulate T-cell activation but also molecules that are involved in leukocyte–endothelial interactions, such as integrins, intercellular adhesion molecules, and chemokine receptors.[295,299] Hence ATG interferes with leukocyte–endothelial interactions that play a role in ischemia-reperfusion injury and can potentially decrease the inflammatory response in patients at risk of delayed graft function. Lastly, Thymoglobulin promotes the induction and proliferation of Treg cells, which play an important role in limiting the immune response to alloantigens.[300,301]

A recent meta-analysis concluded that ATG used as an induction agent prevents acute rejection, but increases the risk of CMV infection, thrombocytopenia, and leukopenia. However, the impact of ATG on death, graft survival, and malignancy are unclear.[302] ATG is also used as a treatment for acute rejection.[291]

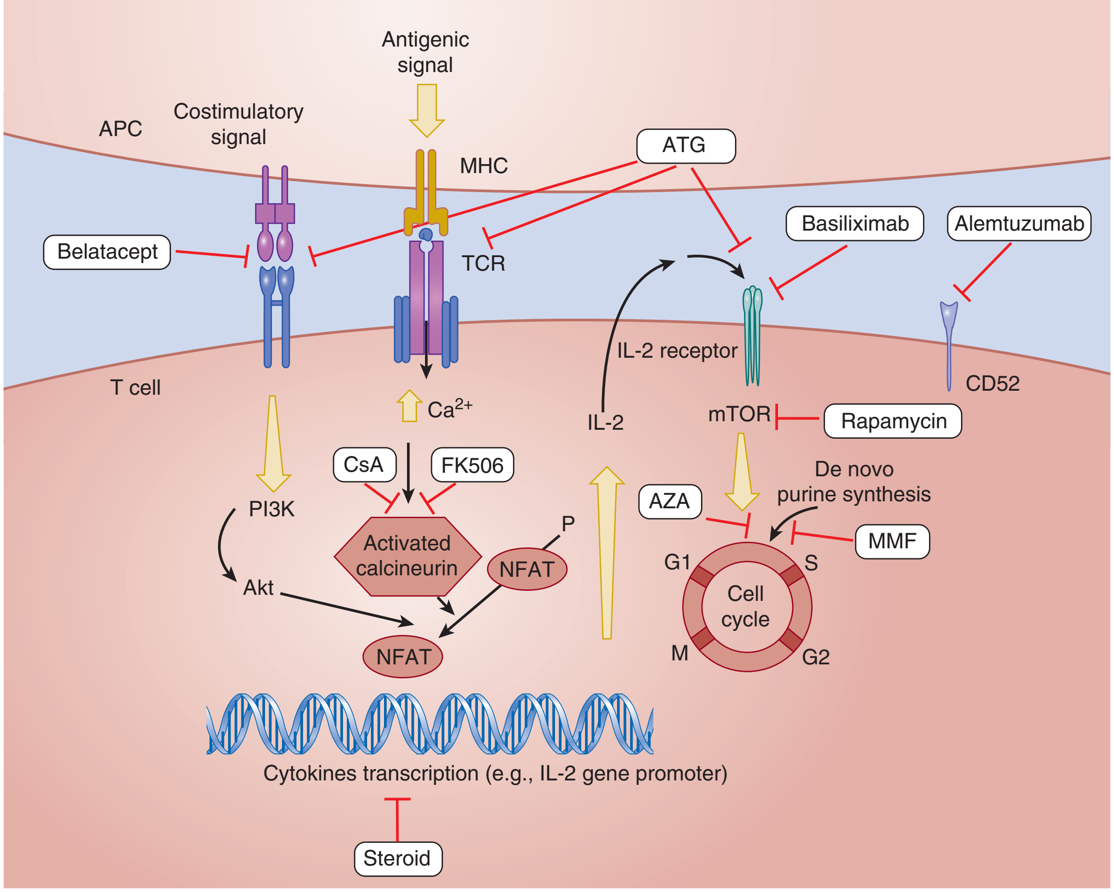

Fig. 69.11 Mechanisms of action of immunosuppressive agents viewed as a function of inhibiting T-cell activation. Signal 1: The calcium-dependent signal induced by T-cell receptor (TCR) stimulation results in calcineurin activation, a process inhibited by cyclosporine (CsA) and tacrolimus. Calcineurin dephosphorylates nuclear factor of activated T cells (NFAT), enabling it to enter the nucleus and bind to the interleukin-2 (IL-2) promoter. Corticosteroids bind to cytoplasmic receptors, enter the nucleus, and inhibit cytokine gene transcription in both T cells and the antigen-presenting cells (APCs). Corticosteroids also inhibit nuclear factorκlight-chain enhancer of activated B cells (NF-κB) activation (not shown). Signal 2: Costimulatory signals are necessary to optimize T-cell IL-2 gene transcription, prevent T-cell energy, and inhibit T-cell apoptosis (belatacept target). Signal 3: IL-2 receptor stimulation induces the cell to enter the cell cycle and proliferate. Signal 3 may be blocked by IL-2 receptor antibodies (basiliximab) or by mammalian target of rapamycin (mTOR) inhibitors (such as rapamycin), which inhibits S6 kinase activation. Following progression into the cell cycle, azathioprine (AZA) and mycophenolate mofetil (MMF) interrupt DNA replication by inhibiting purine synthesis. Antithymocyte globulin (ATG) has multiple targets, including IL-2R, CD3, and CD28, leading to T-cell depletion. Alemtuzumab selectively targets CD52 receptor on T cells. *MHC*, Major histocompatibility complex; *PI3K*, phosphatidylinositol-3-kinase.

MONOCLONAL ANTIBODIES

ANTI-IL-2R ANTIBODIES

In contrast to polyclonal antibodies and alemtuzumab, anti-IL-2R antibodies are nondepleting, chimeric, monoclonal antibodies targeted against CD25, or IL-2R, which is expressed on activated lymphocytes. Since daclizumab was withdrawn from the market because of lack of demand in 2009, basiliximab is currently the only agent in the anti-IL-2R class. Basiliximab is composed of mouse light and heavy chains and a human Fc portion. Activated T cells produce IL-2, which acts in an autocrine fashion on IL-2R to induce T-cell proliferation (Fig. 69.11).[303] Binding to the α subunit of IL-2R, basiliximab competitively inhibits IL-2–induced lymphocyte proliferation, therefore suppressing T-cell activity against the allograft. Internalization of IL-2R follows basiliximab administration and is associated with decreased counts of circulating CD25-expressing T cells.[296,304,305] Basiliximab induction achieves complete IL-2R saturation and T-cell suppression for 4 to 6 weeks.[305]

Anti-IL-2R antibodies are indicated in the prevention, but not the treatment, of rejection. Their use decreases rejection rates and early graft loss when compared with placebo.[306] Although randomized controlled trials have reported that ATG was superior to anti-IL-2R antibodies in the prevention of acute rejection in patients at high risk of delayed graft function or rejection,[307,308] graft survival was similar between the two treatment groups up to 5 years posttransplantation.[309] Despite the fact that severe hyperreactivity reactions have been described, anti-IL-2R antibodies are generally well tolerated.

ALEMTUZUMAB

Alemtuzumab (Campath-1H) is a humanized, rat monoclonal IgG1 antibody directed against CD52. CD52 is a cell surface glycoprotein expressed on both B and T cells and on NK cells, monocytes, macrophages, and DCs. Recent data suggest that CD52 is involved in T-cell costimulation, migration, and adhesion, and may also induce T-reg cells.[310] Alemtuzumab acts though the depletion of B cells, T cells, and other cell lines expressing CD52. First, binding of alemtuzumab to CD52 on the cell surface can activate the complement cascade through the classical pathway, leading to formation of the membrane attack complex and target cell lysis. Second, through their IgG fragment C receptor (FcƴR), NK cells and macrophages can recognize the Fc region of lymphocyte-bound alemtuzumab, favoring the release of perforins and granzymes, and leading to target cell lysis. Lastly, binding of alemtuzumab seems to induce a nonclassical, caspase-independent, apoptotic response in target cells.[311,312]

Alemtuzumab induces a rapid but long-lasting depletion of B and T cells in the peripheral blood and secondary lymphoid organs. After standard induction treatment, reconstitution occurs at a variable rate in kidney transplant patients, with B cells recovering after 12 months and T-cell counts recovering to approximately 50% of baseline after 36 months.[313] Furthermore, alemtuzumab treatment induces an increase in FoxP3+ Treg cells and an anti-inflammatory pattern in the immune system.[314,315] Although alemtuzumab is a humanized antibody, it contains regions originating from a rat IgG2 antibody, and the development of antidrug antibodies has been reported.[316]

Alemtuzumab is used off label as an induction agent for the prevention of rejection and as a rejection treatment in kidney transplant recipients. Systematic reviews show that when administered as an induction agent to prevent rejection, the efficacy of alemtuzumab is similar to that of ATG.[302,317] Nevertheless, some studies have reported an increased risk of formation of *de novo* DSA compared with basiliximab or ATG,[318] as well as an increase in the incidence of acute ABMR.[319–321] Alemtuzumab has also been used as a treatment for acute TCMR in patients with steroid-resistant rejection in observational studies. The efficacy of alemtuzumab appears similar to that of Thymoglobulin in this context,[310] although large randomized controlled trials are lacking.

MAINTENANCE IMMUNOSUPPRESSIVE DRUGS

Maintenance immunosuppressive regimens usually consist of the concomitant administration of 3 agents that act through different pathways in a synergistic fashion, with the goal of avoiding allograft rejection. Although other combinations exist, the combination of tacrolimus, mycophenolate mofetil, and corticosteroids is most frequently used.

CALCINEURIN INHIBITORS

Although tacrolimus and cyclosporine differ in origin, intracellular protein binding, and side effect profiles, they share a common final pathway in terms of mechanism of action: the inhibition of calcineurin. Calcineurin is a serine threonine phosphatase that dephosphorylates the cytosolic component of the nuclear factor of activated T cells (NFAT) (Fig. 69.11) After dephosphorylation, NFAT is translocated from the cytosol to the nucleus, where it forms a complex with other DNA-binding proteins, including FOS and JUN. This complex regulates gene transcription, including the gene for IL-2. During treatment with calcineurin inhibitors, the inhibition of the formation of the NFAT complex has been shown to prevent transcription of the *IL-2* gene, and a similar complex has been shown to regulate *TNF-α* gene transcription. The net result is a reduction in T-cell activation and, indirectly though decreased IL-2 production, decreased T-cell proliferation.

DIFFERENCES BETWEEN TACCOLIMUS AND CYCLOSPORINE

Tacrolimus (FK506) is a macrolide antibiotic produced by fungi that binds to a family of cytosolic proteins, the FK binding proteins. In contrast, cyclosporine is a small cyclic peptide of fungal origin that binds to a family of cytoplasmic molecules termed *cyclophilins*. As mentioned previously, both medication-bound protein complexes then inhibit calcineurin. Tacrolimus is now generally preferred to cyclosporine, as randomized controlled trials have demonstrated superior efficacy in terms of acute rejection prevention and graft function 1 year and 3 years posttransplantation.[322–324] The side effect profiles of tacrolimus and cyclosporine are similar, with a few exceptions. Although hirsutism and gingival hyperplasia are caused by cyclosporine, tacrolimus is associated with alopecia instead. Neurotoxicity, diarrhea, and post-transplant diabetes are more frequent with tacrolimus, whereas dyslipidemia is more commonly seen with cyclosporine.[325] In contrast with cyclosporine, tacrolimus is now available as once-daily formulations.[326]

MYCOPHENOLIC ACID

Mycophenolic acid (MPA) is a fermentation product of *Penicillium brevicompactum* fungi that inhibits the proliferation of human T and B cells.[327] The production of purines, which are subunits of nucleic acids, is necessary for DNA synthesis and cell proliferation. There are 2 routes for the synthesis of nucleotides: the de novo and the salvage pathways. The de novo pathway uses simple molecules, such as CO_2, amino acids, and tetrahydrofolate, to build novel purine molecules. In contrast, the salvage pathway recycles purine bases. While most cells use both pathways to synthesize purines, T and B cells are dependent on the de novo pathway. The main mechanism of action of MPA is the inhibition of inosine-5'-monophosphate dehydrogenase (IMPDH), an enzyme that limits the rate of guanosine nucleotide synthesis through the de novo pathway, thereby limiting DNA synthesis and cell proliferation. IMPDH has two isoforms. Although the type I isoform is expressed in most cell types, the type II isoform is preferentially expressed in T and B cells.[328] MPA inhibits the type II IMPDH isoform 5 times more potently than the type I isoform. The preferential expression of type II IMPDH in lymphocytes and the dependence of lymphocytes on the de novo pathway for purine synthesis may explain why MPA's antiproliferative effect is mostly observed in lymphocytes,[329] although monocytes can also be affected.[330] MPA has also been shown to decrease antibody responses to multiple alloantigens, including CMV[331] and equine proteins found in ATGAM preparations.[332] In addition, it is believed to suppress DSA production,[333] although this can still develop with the use of combinations of tacrolimus and MPA. MPA also suppresses autoantibody production through early inhibition of B-cell proliferation and differentiation,[329] and is now a recognized treatment for autoimmune diseases such as lupus nephritis.[268]

MPA has other mechanisms of action that are relevant in the context of organ transplantation. For instance, MPA can suppress the maturation of DCs and inhibit their capacity to present antigens.[334,335] MPA can also reduce the recruitment of monocytes into sites of graft rejection and inflammation, and induce apoptosis in these cells,[336] hence exerting an anti-inflammatory effect. Finally, MPA reduces the expression of adhesion molecules on endothelial cells[337] and decreases lymphocyte adhesion/infiltration of the allograft.[329]

MPA has superior efficacy in preventing acute rejection episodes and death-censored graft loss when compared with azathioprine,[338] and it is used in combination with calcineurin inhibitors and prednisone in most transplant centers.[256]

AZATHIOPRINE

Azathioprine is a purine analog that is enzymatically converted in vivo to 6-mercaptopurine and other derivatives, which are molecules that function as antimetabolites. After metabolic conversion, it has multiple activities, including incorporation into DNA, inhibition of purine nucleotide synthesis, and alteration of RNA synthesis. The major immunosuppressive effect is thought to be caused by blocking of DNA replication, which prevents lymphocyte proliferation after antigenic stimulation. Although it is useful for inhibiting primary immune responses, azathioprine has little effect on secondary responses or the reversal of acute allograft rejections, which are not dependent on lymphocyte proliferation. Azathioprine also decreases the number of migratory mononuclear and granulocytic cells, while inhibiting proliferation of promyelocytes in the bone marrow. As a result, the number of circulating monocytes capable of differentiating into macrophages is decreased. Among the possible deleterious effects of azathioprine are leukopenia, thrombocytopenia, hepatotoxicity, and an increased risk of neoplasia. Pancreatitis is also a rare but serious side effect of azathioprine. Azathioprine can be used in combination with calcineurin inhibitors in patients who are intolerant to MPA. Azathioprine-treated patients are less likely to experience gastrointestinal disturbances but are more likely to have thrombocytopenia and elevated liver enzymes than those receiving MPA.[338]

MAMMALIAN TARGET OF RAPAMYCIN INHIBITORS

Rapamycin, or sirolimus, is a macrolide antibiotic that blocks T-cell proliferation during the late G1 phase of the cell cycle and before the S phase, resulting in a mid-to-late G1 arrest of the cell cycle.[339] After binding to intracellular FK-binding proteins, rapamycin forms an effector complex that inhibits mTOR. Everolimus (RAD) is a rapamycin derivative that has a similar mode of action to sirolimus. mTOR inhibition prevents phosphorylation of the p70 S6 kinase, which reduces the translation of mRNAs that encode ribosomal proteins and elongation factors, ultimately decreasing protein synthesis.[339] In addition, rapamycin prevents the decline of p27 cyclin-dependent kinase (cdk) inhibitor that typically follows IL-2 stimulation. In turn, this leads to an inhibition of the enzymatic activity of the cdk2-cyclin E complex, a regulator of G1/S transition. In functional studies, activation of T cells by monoclonal antibodies against the TCR was inhibited by either cyclosporine or tacrolimus, but not rapamycin. Conversely, the activation of T cells by exogenous IL-2 plus protein kinase C stimulation with a phorbol ester was inhibited by rapamycin, but not cyclosporine or tacrolimus.[340] Similarly, T-cell activation by monoclonal antibodies against the CD28 costimulatory molecule plus protein kinase costimulation with a phorbol ester was also inhibited by rapamycin, but not cyclosporine or tacrolimus. In contrast, cyclosporine and tacrolimus inhibit an early signal in T-cell activation that is transduced by the TCR signal transduction pathway. Hence rapamycin exerts its effect by interfering with the CD28 costimulatory and IL-2R signal transduction pathways.

Even though rapamycin interacts with the same binding proteins as tacrolimus, there is no competitive inhibition between these drugs, because the binding proteins are present in great excess compared with tacrolimus and rapamycin. The mTOR inhibitors have been shown to be potent inhibitors of vascular endothelial growth factor, which may explain their role in preventing progression of many forms of cancer. mTOR inhibitors have been used in combination with calcineurin inhibitors to replace azathioprine or MPA, as well as in combination with MPA to replace calcineurin inhibitors, which are associated with nephrotoxicity. In a recent meta-analysis, kidney transplant recipients who received mTOR inhibitors were at a lower risk of neoplasia, but at higher risk of death.[341] Whereas conversion from calcineurin to mTOR inhibitors is associated with better short-term graft function,[342] enhanced long-term survival has never been demonstrated. Moreover, retrospective registry cohort studies have suggested an increased risk of graft loss with mTOR inhibitors in combination with, or in replacement of calcineurin inhibitors[343] Additionally, increased de novo DSA have been reported in patients who switched from calcineurin to mTOR inhibitors.[344] mTOR inhibitors are not routinely administered in most transplant centers, but benefit may be derived from their use for selected populations, such as patients at high risk of cancer and those who are intolerant or show toxicity from calcineurin inhibitors. Nevertheless, side effects can be frequent, and may include dyslipidemia, peripheral edema, cytopenia, acne, proteinuria, and oral ulcers.[345]

CORTICOSTEROIDS

Corticosteroids modulate the immune response by regulating gene expression. The steroid molecule enters the cytosol, where it binds to the steroid receptor and induces a conformational change in the latter. The complex then migrates to the nucleus and binds regulatory regions of DNA called glucocorticoid response elements, which regulate the transcription of many genes, including *IL-1*, *IL-2*, *IFN-γ*, *TNF-α*, and *IL-6*. Most immunosuppressive drug regimens include a corticosteroid, such as prednisone, in combination with other immunosuppressive agents. A recent meta-analysis showed that steroid avoidance (use of corticosteroids for less than 14 days posttransplant) or steroid withdrawal (discontinuation of corticosteroids after more than 14 days of use posttransplant) led to an increased risk of acute rejection, while there was no significant effect on graft loss or patient mortality up to 1 year posttransplant.[346] The risk of post-transplant diabetes and CMV infection were similar whether or not steroids were maintained. However, long-term data on the risk and benefits of steroid avoidance or discontinuation are lacking.

BELATACEPT

Belatacept is a fusion protein composed of an Fc fragment of human IgG1 linked to an extracellular domain of cytotoxic T lymphocyte antigen-4 (CTLA-4), which is a molecule expressed on activated T cells.[347] As discussed earlier in this chapter, the CD28 molecule is constitutively expressed on naïve T cells. After binding with B7-1 and B7-2 (also called CD80 and CD86, respectively) on the surface of APCs, CD28 provides a costimulatory signal that is crucial for T-cell activation and proliferation via IL-2 secretion.[348] CD80 and CD86 also bind with CTLA-4, inducing a negative regulatory effect on T-cell activation.[349] The affinity of CD80/CD86 is much greater for CTLA-4 than for CD28.[350] Through its CTLA-4 domain, belatacept binds to CD80/CD86 on the surface of APCs, thus preventing the binding of CD80/CD86 to CD28. This then impedes the CD28-mediated costimulatory signal that is required for T-cell activation.[348]

Belatacept is the first intravenous maintenance immunosuppressive agent that has been marketed. It is not used for the treatment of acute rejection episodes. In de novo transplant recipients receiving standard or extended criteria donors, use of belatacept (versus cyclosporine) in combination with mycophenolate mofetil, prednisone, and basiliximab induction has resulted in higher graft function 1 year posttransplantation, although the rate of acute rejection episodes was higher.[351] Follow-up studies after 7 years have suggested that the benefit seen in graft function preservation may be maintained long term.[352,353] Cases of posttransplant lymphoproliferative diseases have been observed in patients with belatacept, especially in transplant recipients who were EBV-seronegative before transplantation, leading the US Food and Drug Administration (FDA) to issue a black box warning against the use of belatacept in EBV-seronegative recipients.[354] Conversion from calcineurin inhibitors to belatacept has resulted in an improvement in graft function up to 3 years after conversion.[355] Hence belatacept can be used as an alternative for recipients who are intolerant or have nephrotoxicity associated with calcineurin inhibitors.[55]

OTHER IMMUNOSUPPRESSIVE AGENTS USED IN THE FIELD OF KIDNEY TRANSPLANTATION

RITUXIMAB

Rituximab is a chimeric murine/human monoclonal antibody directed against CD20, a transmembrane protein expressed by B cells but not plasma cells or memory B cells.[296] Rituximab strongly depletes peripheral B-cell levels via complement-dependent cytotoxicity, ADCC, and caspase-dependent apoptosis.[356] Rituximab has been studied as an induction agent, as well as for the treatment of acute and chronic ABMR, but clinical benefits remain uncertain at present time.[302,357–359]

BORTEZOMIB

Bortezomib is a reversible inhibitor of the 26S proteasome, a multicatalytic enzyme complex localized in the cell nucleus and cytosol. Since the proteasome is the principal pathway by which cellular proteins are degraded,[360] proteasome inhibition leads to the accumulation of misfolded proteins within the endoplasmic reticulum, leading to apoptosis. Due to their high rate of antibody synthesis, plasma cells are particularly sensitive to proteasome inhibition, and bortezomib has been shown to deplete plasma cell levels.[361]

Although bortezomib has successfully been used in desensitization protocols that also included rituximab and intravenous immunoglobulins in some small case series,[362,363] its benefits remain controversial.[364] Multiple case reports and case series have reported reversal of acute ABMR using bortezomib,[365–369] although its beneficial effect seems to be limited to ABMR episodes occurring in the early posttransplant period.[214] The impact of bortezomib on late ABMR is currently being investigated in a randomized controlled trial.[370]

ECULIZUMAB

Eculizumab is a humanized monoclonal antibody that targets the complement C5a component and prevents its cleavage, thereby inhibiting the formation of the membrane attack complex and preventing complement-mediated injury.[296] Eculizumab prevents the recurrence of atypical hemolytic uremic syndrome after transplantation,[371,372] and has reversed acute ABMR in case reports and small case series,[296] sometimes in combination with splenectomy.[373] Meningococcal vaccination is recommended at least 2 weeks before administering the first dose of eculizumab if the risk of delaying therapy is not judged prohibitive. The use of antibiotic prophylaxis (2 weeks of ciprofloxacin followed by amoxicillin, for instance) can be considered for the duration of treatment with eculizumab, especially when vaccination has not taken place in a timely fashion.[374]

TOCILIZUMAB

Tocilizumab is a humanized monoclonal antibody directed at the soluble and membrane forms of IL-6R. IL-6 is a pleiotropic cytokine that is not expressed in healthy individuals but is rapidly synthesized during infection and tissue injury. Innate immune cells, such as monocytes and macrophages, release IL-6 after recognizing PAMPs or DAMPs. IL-6 stimulates B cells, promotes antibody synthesis and Tfh cell populations, and preferentially induces proinflammatory Th17 cell differentiation at the expense of Treg-cell induction.[296,375] Tocilizumab is FDA-approved for the treatment of severe rheumatoid arthritis and juvenile arthritis. Although not studied on a large scale in the field of transplantation, tocilizumab has shown promising results in desensitization protocols,[376,377] as well as for the treatment of chronic ABMR.[378]

CONCLUDING REMARKS

The field of transplantation immunobiology has been the topic of abundant research efforts in the last decades. In this chapter, we have provided an overview of alloantigen recognition, HLA antigen/antibodies, and mechanisms of rejection and immunosuppressive agents focusing on their relevance to current clinical practice in kidney transplantation. Immunosuppression remains a double-edged sword for kidney transplant patients. Although current immunosuppressive regimen has yielded excellent results in terms of acute rejection prevention and increased short-term graft survival, the risk of infection and cancer remain problematic. While xenotransplantation and genetically engineered grafts are not ready to move to the bedside,[379] the induction of donor-specific tolerance by mixed chimerism[380,381] and regulatory cell therapy[382] are currently being studied in humans. Such initiatives represent future perspectives to decreased adverse effects of immunosuppressants while preserving long-term graft function.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. A 34-year-old female undergoes evaluation for a living donor kidney transplantation. She has had two pregnancies in the past. She has no detectable anti-HLA antibody to her donor on her most recent serum. Her current serum flow cytometry crossmatch is negative for B and T cells. However, she has a positive historical flow cytometry crossmatch on both donor B and T cells. How would her immunological risk best be defined?
   a. There is no increase in immunological risk as the current flow cytometry crossmatch is negative.
   b. The increased immunological risk represents an absolute contraindication to transplantation and would be regarded as such in all transplant centers.
   c. The immunological risk in increased. While some centers may choose to forego transplantation from this donor, others may choose to proceed and adapt immunosuppression and follow-up accordingly.

   Answer: c

   Rationale: A positive flow cytometry crossmatch on historical (remote) serum is a marker of increased immunological risk even when the current (most recent serum) flow cytometry crossmatch is negative, as a memory B-cell response can occur on re-exposure to the HLA antigen to which someone was previously sensitized, in this case probably through pregnancies. Depending on the likelihood of finding another donor for this candidate in a timely fashion and on local practice, some centers may forego transplantation using this donor, while others may accept it but give higher levels of immunosuppression.

2. A 55-year-old female received a kidney transplantation 5 years ago. A kidney biopsy is performed because of slowly rising creatinine levels and proteinuria. The biopsy shows double contours in glomerular capillaries. What is the most likely diagnosis?
   a. Chronic antibody-mediated rejection
   b. Cell-mediated chronic rejection
   c. Acute antibody-mediated rejection

   Answer: a

   Rationale: Double contours in glomerular capillaries is a marker of antibody-mediated, chronic endothelial injury and remodeling. Here, the biopsy could also show multilayering of peritubular capillary membrane, again a marker of microcirculatory chronic endothelial injury. In chronic cell-mediated rejection, vascular remodeling with mononuclear cell infiltration is typically observed in small-sized arteries. Double contours are generally not observed when the antibody-mediated insult is acute.

3. A 45-year-old male who received a kidney transplantation 15 years ago has increasing serum creatinine in the absence of proteinuria. He has a history of corticosensitive T-cell–mediated rejection 3 months posttransplant. The questionnaire is suspicious for nonadherence to mycophenolate mofetil, but the patient reports that he is taking his calcineurin inhibitor as prescribed. Which of the following is true?
   a. The biopsy is most likely to show calcineurin inhibitor toxicity. After 15 years of transplantation, the risk of rejection is very low when mycophenolate mofetil is discontinued.
   b. Rejection is a likely diagnosis given the history of nonadherence and a previous history of rejection, as mycophenolate mofetil inhibits both T- and B-cell proliferation.
   c. As mycophenolate mofetil is an inhibitor of the cosignal between antigen-presenting cells and T cells, rejection is very likely.

   Answer: b

   Rationale: Although calcineurin inhibitor toxicity could be found on the biopsy, a history of nonadherence to mycophenolate mofetil should still be considered suspicious for rejection, as allorecognition persists through the years. Mycophenolate mofetil inhibits T- and B-cell proliferation by inhibiting IMPDH, a key enzyme in the *de novo* pathway of purine synthesis. Belatacept is an inhibitor of the cosignal between antigen presenting cells and T cells, but not mycophenolate mofetil.